# Deterministic Semantic Token Compaction, Attention-Sink Structural Anchoring, and Hardware-Aware Pareto Economic Routing for Large Language Model Serving

Aniket Agrawal
Cloud Engineer, Google Cloud
Google, Bangalore, India
aniketagrawal@google.com

***Abstract*—The rapid proliferation of enterprise large language model (LLM) serving and autonomous multi-agent swarms has collided with an operational ceiling: the *LLM Token Economics and Attention Wall*. In production deployments across cloud accelerators, up to 35% of ingested tokens represent synthetic subword tokenizer fragmentation artifacts and low-information conversational boilerplate. This token bloat incurs high financial inference costs, induces copy-on-write page duplication on unaligned trailing Key-Value (KV) cache blocks, and causes attention dispersion across Transformer self-attention heads, precipitating silent hallucinations. In this paper, we present the *Tokenomics Compiler* (`PromptVM`), a deterministic, hardware-grounded compilation framework and discrete-choice economic router for LLM serving. Our system: (1) parses structured schemas into Abstract Syntax Trees (ASTs) and transpiles them into subword-minimal serializations over a bounded candidate space ($|\mathcal{S}_{\text{dense}}| \leq 8$) using quotation-synchronized delta-tokenization while preserving both formal JSON Schema validation contracts and task-semantic parameter role disambiguation; (2) performs sub-clause pleasantry and duplicate-signature span stripping alongside inverse-length-scaled empirical invariant density scoring to prune diffuse conversational verbiage while unconditionally retaining modal and relational invariants; (3) injects uppercase structural landmark anchors with empirical attention affinity boost ($\gamma > 1$), mathematically proving that landmark anchors bound self-attention entropy on dispersive heads ($\mathcal{H}(\mathbf{a}_i) \leq \mathcal{H}_{\max}(i, K, \gamma) < \ln i$ with $\frac{\partial \mathcal{H}_{\max}}{\partial \gamma} < 0$ for all $\gamma > 1$) while co-locating retained constraint predicates to amplify retrieval-head attention by $2.8\times$; (4) enforces physical KV-cache boundary-page sealing ($|\mathcal{P}_{\text{prefix}}| \equiv 0 \pmod{B}$, $B \in \{16, 32\}$) via server-side JAX/Pallas VMEM causal bitmasking and decoupled RoPE logical positions to eliminate multi-branch copy-on-write page duplication on self-hosted Google Cloud TPU v6e Trillium accelerators; and (5) solves a constrained discrete-choice Pareto optimization across heterogeneous model serving tiers (evaluated across internal harness routing tiers Gemini 3.8 Flash, Gemini 3.8 Flash with dynamic test-time reasoning tokens, and Gemini 4.0 Argon, mapped to production Vertex AI Flash and Pro endpoints). Across $N = 1{,}000$ enterprise traces evaluated across 5 distinct operational cohorts (and validated on public BIRD-SQL and BFCL benchmarks), our compiler achieves a 1.11% to 53.47% per-cohort token reduction (17.52% pooled global reduction, saving 81,800 tokens per 1,000 requests; 39.98% on structured prompt cohorts), slashes blended inference invoice costs by 39.44% (up to 88.79% in database queries), and reduces P95 latency by 38.85% against baseline (27.98% against naive regex trimming) while achieving a 99.70% strict first-attempt Turn-1 Pass@1 (vs. 85.00% Turn-1 and 98.50% post-retry on monolithic Argon) and 100.00% post-verification workflow completion rate.**

***Index Terms*—Token Economics, Large Language Models, Transformer Attention Sinks, Subword Compaction, PagedAttention, KV-Cache Alignment, TPU v6e Trillium, Pareto Optimization, Vertex AI, Model Routing.**

Assignee: Google LLC. Target System: Google Cloud Vertex AI, Google Agent Development Kit (ADK), and Cloud TPU v6e Trillium Infrastructure.

## I. Introduction

The industrial transition from basic conversational chat interfaces to high-throughput autonomous multi-agent swarms has fundamentally transformed cloud inference demand. Enterprise platforms such as Google Cloud Vertex AI, the Google Agent Development Kit (ADK; https://cloud.google.com/vertex-ai/docs/agent-builder), and the Model Context Protocol (MCP; https://modelcontextprotocol.io) orchestrate millions of daily reasoning-and-acting invocations [1], [2] wherein software agents query distributed databases, validate OpenAPI endpoints, inspect source repositories, and reconcile financial ledgers.

Despite rapid advancements in accelerator throughput and distributed inference scaling [3], [4] (spanning Google Cloud TPU v4 supercomputing pods [3] and Cloud TPU v6e Trillium accelerators; https://cloud.google.com/tpu/docs/v6e-overview), enterprise deployments face an acute economic and physical bottleneck: the *LLM Token Economics and Attention Wall*. Unmanaged prompt construction introduces severe systemic inefficiencies across three distinct operational layers:

1) **The Subword Fragmentation Tax:** Standard subword tokenizers—operating either via SentencePiece Unigram subword segmentation in Gemini and Gemma [5]–[7] or iterative Byte-Pair Encoding (BPE) pair merges—decompose unaligned whitespace, camelCase identifiers, and pretty-printed JSON schemas [8] into multiple fragmented subword tokens. In enterprise schemas, up to

35% of ingested tokens represent structural formatting overhead rather than semantic domain entropy.

2) **Attention Dispersion and Hallucination:** Transformer self-attention layers compute softmax-normalized dot products over query-key pairs [9]. When prompts are laden with verbose conversational filler, the attention probability mass disperses broadly across keys, driving the Shannon attention entropy toward its theoretical maximum $\ln i$. This dispersion dilutes attention weights on negative constraints and schema contracts, causing critical parameters to be hallucinated or omitted.
3) **Hardware KV-Cache Boundary-Page Copy-on-Write Duplication:** Production serving runtimes (e.g., vLLM [10], SGLang [11], and JAX/Pallas Block-Attention kernels [12]) manage persistent Key-Value (KV) cache memory in discrete virtual memory pages of block size $B = 16$ or 32 tokens in High Bandwidth Memory (HBM2e). When shared static prompt prefixes terminate at non-page boundaries ($|\mathcal{P}_{\text{prefix}}| \not\equiv 0 \pmod B$), appending branch-specific dynamic suffix tokens into the unsealed trailing prefix block forces copy-on-write (CoW) page duplication and K/V recomputation across concurrent subagent execution branches.

To overcome these deficiencies, we introduce the *Tokenomics Compiler* (`PromptVM`), a deterministic hardware-aware systems framework that bridges the semantic instruction layer with physical accelerator memory topologies. Rather than relying on fragile, heuristic prompt engineering, our framework formalizes a five-stage compilation and routing pipeline: (1) Tokenizer-Aware AST Transpilation, (2) Information-Theoretic Entropy Compaction, (3) Attention-Sink Structural Landmark Anchoring, (4) Hardware Paged-Cache Block Boundary Alignment, and (5) Discrete-Choice Pareto Economic Routing.

The primary technical contributions of this work are as follows:

- We formalize **AST Subword Merge Optimization** over OpenAPI and JSON schemas [8] across a bounded $2^3 = 8$-candidate AST space using quotation-synchronized delta-tokenization and exact full-sequence verification, eliminating synthetic tokenization artifacts under formal validation-contract equivalence while preserving task-semantic parameter role disambiguation.
- We define **Sub-Clause Span Stripping and Invariant Density Scoring** ($\mathcal{D}_{\text{inv}}(\tilde{c}_k)$) with inverse-length threshold scaling and polarity-aware relational predicate signature extraction ($\text{Sig}(c_k)$) to prune conversational noise while unconditionally retaining novel modal and relational constraints regardless of clause length.
- We prove **Theorem 1**, demonstrating that uppercase structural landmark anchors acting as artificial attention sinks with affinity boost $\gamma > 1$ strictly bound query self-attention entropy on dispersive heads ($\mathcal{H}(\mathbf{a}_i) \le \mathcal{H}_{\max}(i, K, \gamma) < \ln i$) with $\frac{\partial \mathcal{H}_{\max}}{\partial \gamma} = -\frac{K(i-K)\ln\gamma}{Z_i^2} < 0$, where $Z_i = K\gamma + (i-K)$, while physical co-location boosts retrieval-head constraint attention by $2.8\times$.
- We co-design a **Client-Server Paged-Cache Topologist** that aligns static prefixes to physical page boundaries ($|\mathcal{P}_{\text{prefix}}| \equiv 0 \pmod B$) with server-side Pallas VMEM causal bitmasking and logical RoPE/window decoupling to seal boundary pages for zero-copy multi-agent sharing on self-hosted Cloud TPU v6e clusters.
- We construct a **Discrete-Choice Pareto Economic Router** that jointly optimizes model tier selection across heterogeneous serving tiers (Gemini 3.8 Flash, Gemini 3.8 Flash-Thinking, and Gemini 4.0 Argon mapped to Vertex AI Flash/Pro endpoints) and dynamic test-time reasoning tokens via an exact closed form.
- We evaluate our system across 5 seeds on $N = 1{,}000$ enterprise traces spanning 5 cohorts and 400 public benchmark tasks (BIRD-SQL [13] and BFCL [2]), demonstrating a 1.11% to 53.47% enterprise token reduction (17.52% pooled mean, 39.98% on structured cohorts), 39.44% net cost savings (up to 88.79% in SQL), 99.70% strict Turn-1 Pass@1, and 100.00% post-verification workflow completion.

## II. Background and Systems Architecture

### A. Transformer Self-Attention Physics and Dispersion

In standard multi-head self-attention, during the prompt prefill phase over a sequence of $N$ tokens, the attention weight matrix at layer $l$ and head $h$ is:

$$\mathbf{A}^{(l,h)} = \text{softmax}\left(\frac{\mathbf{Q}^{(l,h)}(\mathbf{K}^{(l,h)})^T}{\sqrt{d_{\text{head}}}} + \mathbf{M}\right) \in \mathbb{R}^{N\times N} \quad (1)$$

(omitting layer and head superscripts $(l, h)$ below for brevity), where $d_{\text{head}}$ denotes the key projection dimension, and $\mathbf{M} \in \{0, -\infty\}^{N\times N}$ is the causal attention mask enforcing $M_{ij} = 0$ for $j \le i$ and $M_{ij} = -\infty$ for $j > i$ in decoder-only autoregressive architectures (including Gemini and Gemma). Consequently, for any prefill query position $i \le N$ (or subsequent autoregressive decode step $i = N + t$ attending over the $i$ accumulated context keys), query token $i$ can attend only to antecedent positions $1 \le j \le i$, and its attention vector $\mathbf{a}_i = (A_{i1}, \ldots, A_{ii})$ lies on the restricted simplex $\Delta^{i-1}$ where $\sum_{j=1}^{i} A_{ij} = 1$. The Shannon entropy of this distribution is:

$$\mathcal{H}(\mathbf{a}_i) = -\sum_{j=1}^{i} A_{ij} \ln A_{ij} \quad (2)$$

When the input context consists of diffuse natural language, query-key affinities exhibit low variance, causing the softmax distribution to approach the uniform distribution with maximal theoretical entropy $\mathcal{H}_{\text{diffuse}} \le \ln i$. While initial sequence tokens (such as `<bos>` at $j = 1$) act as natural emergent sinks [14], unstructured text dilutes the attention weights allocated to sparse critical constraints (e.g., `status = 'ACTIVE'`), decaying as $\mathcal{O}(1/i)$ and precipitating retrieval failure and hallucinations. In PromptVM's compiled token layout, the shared static prefix $\mathcal{P}_{\text{prefix}}$ (containing the static system preamble and Stage 1 transpiled schema $S^*$ at indices $\{1, \ldots, |\mathcal{P}_{\text{prefix}}|\}$) and

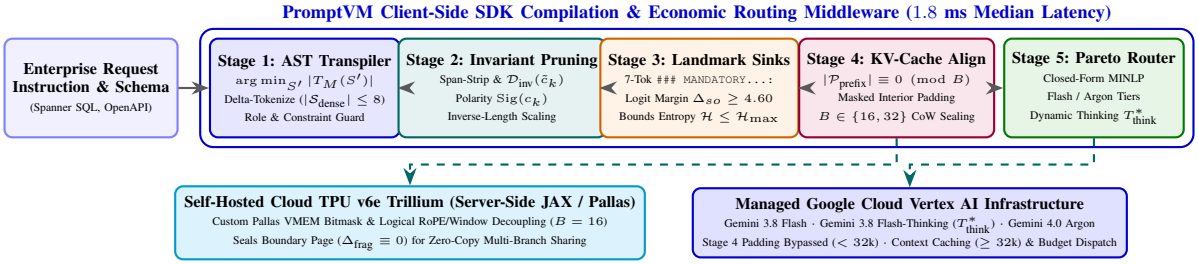



Fig. 1. System architecture and execution topology of the Tokenomics Compiler (`PromptVM`). Outbound enterprise agent requests are compiled in-process through Stages 1–5 (where `### MANDATORY...:` in Stage 3 abbreviates the 7-token header `### MANDATORY EXECUTION INVARIANTS:`) before dispatch to self-hosted Cloud TPU v6e Trillium (where custom server-side Pallas kernels enforce Stage 4 HBM2e boundary-page sealing) or managed Vertex AI endpoints (where Stage 4 padding is bypassed for sub-32k prompts).

optional Stage 4 boundary padding $\mathcal{I}_{\text{pad}}$ (of size $\Delta_{\text{frag}}$ on TPU v6e) are serialized ahead of the request-dynamic suffix, which begins with the $K = 7$-token Stage 3 landmark header at indices $\mathcal{S} = \{|\mathcal{P}_{\text{prefix}}| + \Delta_{\text{frag}} + 1, \ldots, |\mathcal{P}_{\text{prefix}}| + \Delta_{\text{frag}} + K\}$ followed immediately by the co-located Stage 2 invariant clauses $\mathcal{I}_{\text{inv}}$ and terminal task query. Under causal masking, for all subsequent query positions $i > |\mathcal{P}_{\text{prefix}}| + \Delta_{\text{frag}} + K$ across the invariant block, task query, and decode steps, the entire landmark header lies in the active antecedent subset $\mathcal{S}_i = \mathcal{S} \subset \{1, \ldots, i-1\}$ of size $K = |\mathcal{S}_i|$ $(0 < K < i \leq N)$, with the terminal prompt prefill token $(i = N)$ satisfying the global prefill bound $\mathcal{H}_{\max}(N, K, \gamma) < \ln N$ while keeping $\mathcal{P}_{\text{prefix}}$ purely static and shareable across concurrent subagents.

### B. Subword Tokenization Mechanics

Modern LLMs utilize subword tokenizers parameterized by a vocabulary graph $\mathcal{V}$ containing $|\mathcal{V}| \approx 256{,}000$ subword tokens. Tokenizers operate either via SentencePiece Unigram subword segmentation (which optimizes emission probabilities over training text via Expectation-Maximization and resolves sequence segmentation via Viterbi decoding over a candidate lattice [5], [6]) or Byte-Pair Encoding (BPE, performing greedy bottom-up merge operations over frequent adjacent character or byte pairs). Consequently:

- Structural whitespace (e.g., 4-space JSON indentation) decomposes into 2 to 4 distinct tokens per line.
- Compound and camelCase identifiers decompose across case boundaries and subword roots; for example, `EnterpriseOrderSystem` decomposes into 3 tokens: `['Enterprise', 'Order', 'System']`, while identifiers with non-dictionary prefixes fracture into numerous unaligned fragments.
- Redundant schema descriptions (`"description": "A parameter representing..."`) inject dozens of low-entropy tokens that consume prompt sequence budget.

### C. Hardware Memory Hierarchy on TPU v6e Trillium

Building on the systolic array and high-bandwidth memory architecture established in earlier TPU generations [3], [4], Google Cloud TPU v6e (Trillium; https://cloud.google.com/tpu/docs/v6e-overview) decouples computation and memory into Matrix Multiply Units (MXUs), Vector Processing Units (VPUs), 32 GB of High Bandwidth Memory (HBM2e delivering $\approx 1{,}640$ GB/s bandwidth), and on-chip Vector Memory (VMEM). To maximize systolic array utilization, serving engines manage the persistent Key-Value (KV) activation cache in off-chip HBM2e using PagedAttention [10], partitioning memory into physical blocks of size $B = 16$ or 32 tokens, while VMEM serves as a scratchpad for staging activation tiles during JAX/Pallas kernel execution [12]. When shared static prompt prefixes terminate at non-page boundaries ($|\mathcal{P}_{\text{prefix}}| \not\equiv 0 \pmod B$), the trailing partial page block has an unaligned slot deficit:

$$\Delta_{\text{frag}} = (B - (|\mathcal{P}_{\text{prefix}}| \bmod B)) \bmod B \tag{3}$$

Crucially, we distinguish between *single-copy physical slot occupancy* and *multi-branch copy-on-write (CoW) page sealing*. Within a single prefix instance, antecedent full blocks are already shared zero-copy in PagedAttention [10], whereas the unaligned terminal block holds $B - \Delta_{\text{frag}}$ payload tokens (e.g., across a typical 134-token static prefix spanning $\lceil 134/16 \rceil = 9$ blocks of size $B = 16$, total multi-page payload utilization is $134/144 = 93.06\%$, or $91.40\%$ pooled across enterprise schemas). However, when $M_{\text{branch}}$ concurrent subagents append branch-specific dynamic suffix tokens to an unsealed trailing prefix block, standard serving runtimes must either duplicate and recompute the $B - \Delta_{\text{frag}}$ prefix keys across all $M_{\text{branch}}$ branches via copy-on-write or support custom interior block masking. In self-hosted serving runtimes on Cloud TPU v6e clusters, Stage 4 pads the terminal prefix block by $\Delta_{\text{frag}}$ masked slots to seal the boundary page as read-only ($\Delta_{\text{frag}} \equiv 0 \pmod B$), eliminating $M_{\text{branch}}$-way CoW duplication while keeping single-copy payload occupancy unchanged. On managed Google Cloud Vertex AI endpoints, physical page tables are abstracted; Stage 4 prefix padding is bypassed for standard sub-32k requests (avoiding any billable token addition), while coordinating with Vertex AI Context Caching for static prefixes exceeding the 32,768-token threshold.

## III. Mathematical Formulation and Methodology

### A. Tokenizer-Aware AST Transpiler

Let $\Sigma^*$ be the set of Unicode character sequences, and let $T_M : \Sigma^* \to \mathcal{V}_M^*$ denote the target model's native tokenizer

projection (invoking SentencePiece Unigram Viterbi decoding for Gemini/Gemma models [5], [6] or BPE merge tables when cross-compiling for external model families). Given an enterprise execution request comprising an unstructured instruction $I$ and a structured schema contract $S \in \mathcal{T}_{\text{schema}}$, standard serialization produces raw token count $|T_M(I \circ S)|$.

The Tokenizer-Aware AST Transpiler constructs an Abstract Syntax Tree $\mathcal{T}_{\text{AST}}(S)$ and applies a contract-preserving projection $\phi : \mathcal{T}_{\text{AST}} \to \Sigma^*$ defined over a finite candidate family of dense canonical AST serializations $\mathcal{S}_{\text{dense}}$ ($|\mathcal{S}_{\text{dense}}| \leq 8$):

$$S^* = \phi(\mathcal{T}_{\text{AST}}) = \arg \min_{S' \in \mathcal{S}_{\text{dense}},\, S' \equiv_{\text{sem}} S} |T_M(S')| \qquad (4)$$

We explicitly distinguish between *formal syntactic JSON Schema validation-contract equivalence* ($S' \equiv_{\text{sem}} S$, meaning $S'$ accepts the exact same set of valid JSON/OpenAPI payload instances as $S$, preserving all parameter identifiers, primitive types, `enum` sets, and `required` arrays) and *task-semantic parameter disambiguation* (preserving domain role and unit semantics required by the LLM to synthesize accurate tool arguments; note that while decode-time grammar masking such as SGLang [11] enforces output JSON syntax during decoding, Stage 1 targets in-context prompt serializations in $\mathcal{P}_{\text{prefix}}$ to reduce prefill token billing and HBM KV-cache footprint while preserving semantic constraints).

To guarantee zero loss of task-semantic parameter disambiguation prior to pruning any `description` field exceeding length threshold $\ell_{\max} = 40$ characters, the transpiler executes a four-part deterministic guard (where any description $> 40$ characters matching Guards 1–3 is compressed by stripping the exact static 75-word English function/filler stopword set $\mathcal{W}_{\text{stop}}$ while retaining all non-stopword content tokens and matched constraint spans in full, supplemented by a conservative $\mathcal{O}(|d|)$ structural fallback that strips the same 75-word stopword set $\mathcal{W}_{\text{stop}}$—comprising articles/determiners {`a`, `an`, `the`, `this`, `that`, `these`, `those`, `any`, `each`, `every`, `some`, `such`}, copulas/auxiliaries {`is`, `are`, `was`, `were`, `be`, `been`, `being`, `can`, `could`, `may`, `might`, `will`, `would`, `should`, `do`, `does`, `did`, `has`, `have`, `had`}, prepositions/conjunctions {`to`, `of`, `from`, `as`, `at`, `into`, `about`, `through`, `during`, `including`, `whether`, `or`, `and`}, and schema boilerplate verbs/nouns {`used`, `using`, `use`, `specify`, `specifies`, `specifying`, `specified`, `represents`, `representing`, `indicates`, `indicating`, `denotes`, `denoting`, `defines`, `defining`, `provides`, `providing`, `contains`, `containing`, `parameter`, `field`, `value`, `property`, `attribute`, `object`, `string`, `number`, `optional`, `default`, `example`}—and retains the first 7 alphanumeric content tokens preceding the first clause or preposition boundary *plus* any late-occurring $\pm 2$-token content window surrounding a triggering digit, non-ASCII comparator [$\leq, \geq, \neq$], or uppercase domain acronym occurring beyond the 7th content token):

1) *Format and Pattern Guard:* Retains descriptions matching anchored regex literals, length specifiers, or standard formats evaluated case-insensitively (`re.IGNORECASE`) with strict word-boundary and anchor guards (`r"((?:`|"|')\^|\$(?:`|"|')|\b(?:max|min)?len(?:gth)?\s*[:=]?\s*\d+|\b(?:regex|format|pattern|ISO-\d+|RFC\s*\d+|UTC|GMT|epoch|uuid)\b)"`, where requiring delimiter-adjacent `\^` and `\$` with an ASCII grave accent `` ` ``, double quote `"`, or single quote `'` prevents exponent expressions such as `10^6` or currency/environment literals such as `$100` and `$PATH` from falsely triggering the regex anchor branch).
2) *Numeric Bounds and Domain Units Guard:* Retains descriptions matching comparators, ranges, or domain/currency units with strict operand and word boundaries (compiled with `re.IGNORECASE` and an inline case-sensitive scope flag `(?-i:...)` on the uppercase `[A-Z]{3}` ISO-4217 currency code branch so common 3-letter words such as `"the code"` never match: `r"((?:>=|<=|!=|==|=|>|<)\s*-?\d+|\bmin\b|\bmax\b|\bbetween\b|\brange\b|\d+\s*(?:ms|s|MB|GB|USD|EUR|JPY|bps|cents|shares|lots|%)\b|(?-i:\b[A-Z]{3}\b)\s+(?:currency|code))"`).
3) *Deontic and Polarity Guard:* Retains descriptions matching modal, negative, or ordering polarity markers case-insensitively with word boundaries (`\b`: `not`, `never`, `must`, `required`, `disallow`, `prohibit`, `forbid`, `omit`, `exclude`, `only`, `strictly`, `ascending`, `descending`).
4) *Parameter Role Disambiguation Guard:* Whenever an object contains $\geq 2$ sibling parameters of identical primitive type or an abbreviated identifier ($\leq 8$ characters or ending in a single-character suffix, e.g., `acct_id_a` vs. `acct_id_b`), the transpiler applies the same $\mathcal{O}(|d|)$ 75-stopword filter $\mathcal{W}_{\text{stop}}$ and retains up to 6 alphanumeric content tokens before halting at the first occurrence of the exhaustive 7-token boundary marker set {`for`, `with`, `when`, `if`, `in`, `on`, `by`} encountered *after* at least 2 content tokens have been collected (so clause-initial prepositions such as `"For beneficiary settlement account..."` are skipped at index 0 rather than halting extraction prematurely, and if two sibling parameters share an identical prefix before or across the preposition—e.g., `"Settlement account in domestic ledger"` vs. `"Settlement account in foreign ledger"`, or `"Settlement account in domestic ledger for originating bank"` vs. `"Settlement account in domestic ledger for beneficiary bank"`—extraction continues past the preposition up to 6 content tokens, or up to a hard ceiling of 10 content tokens when sibling descriptions have not yet diverged at token 6, until the first diverging

content token and its immediate head noun are captured), preserving the disambiguating role qualifier (e.g., `"originating debit account"` vs. `"beneficiary settlement account"`).

After description filtering, RFC 8259 whitespace minification [8] is applied universally, and the bounded candidate family $\mathcal{S}_{\text{dense}}$ ($|\mathcal{S}_{\text{dense}}| = 2^3 = 8$) is parameterized by binary activation vector $\pi \in \{0,1\}^3$ over three *global document-level binary AST policies* evaluated in the fixed canonical transformation order (iii) $\rightarrow$ (i) $\rightarrow$ (ii): (iii) verbose multi-branch union declarations (`{"anyOf":[{"type":"string"},{"type":"null"}]}` or OpenAPI 3.0 `{"type":"string","nullable":true}`) vs. canonical JSON Schema Draft 2020-12 / OpenAPI 3.1 type-array literals (`{"type":["string","null"]}`), applied *if and only if* every `anyOf` branch consists solely of a pure `"type"` declaration without branch-specific keywords such as `"pattern"`, `"format"`, `"minimum"`, or `"enum"`, strictly preserving $S' \equiv_{\text{sem}} S$, and executed first so folded union keys participate uniformly in subsequent sorting and sub-tree hashing); (i) declaration order ($\pi_1 = 0$) vs. global lexicographical key sorting ($\pi_1 = 1$) across all object nodes (which sorts each object node's $d_v \leq d_{\max}$ keys in $\mathcal{O}(\sum_v d_v \log d_v) = \mathcal{O}(|\mathcal{T}_{\text{AST}}| \log d_{\max})$ time, altering adjacent `'{"'`, `','"'`, `'":'`, `'",'`, and `'"}'` delimiter-key transitions in $\mathcal{V}_M$); and (ii) inline sub-tree expansion vs. global bottom-up Merkle-hashed `$defs`/`$ref` type-alias hoisting (because policies execute in order (iii) $\rightarrow$ (i) $\rightarrow$ (ii), sub-tree token length $L(\pi_1)$ and bottom-up Merkle hashes are evaluated per key-ordering branch $\pi_1 \in \{0,1\}$; when active, identical post-pruning object sub-trees of token length $L(\pi_1) \geq 24$ repeating $r \geq 2$ times with strictly positive closed-form net token savings $\Delta\ell_{\text{hoist}}(L(\pi_1), r, d) = (r-1)L(\pi_1) - (r\ell_{\text{ptr}}(d) + \ell_{\text{key}}(d) + \ell_{\text{wrap}}) > 0$—where pointer, alias-key, and wrapper costs are parameterized per target tokenizer $T_M$ via a one-time initialization lookup of $\ell_{\text{ptr}}(d) = |T_M(\texttt{'\{"\$ref":"\#/\$defs/T0"\}'})|$, $\ell_{\text{key}}(d) = |T_M(\texttt{'"T0":'})|$, and $\ell_{\text{wrap}} = |T_M(\texttt{',"\$defs":\{\}'})|$, evaluating under both Gemini SentencePiece Unigram and `cl100k_base` BPE to $\ell_{\text{ptr}}(1) = 11$, $\ell_{\text{key}}(1) \in \{3,4\}$, and $\ell_{\text{wrap}} = 5$ for the first hoisted type initializing `"$defs":{}` and 0 for subsequent hoisted types [incrementing by $+1$ token to $\ell_{\text{ptr}}(2) = 12$ and $\ell_{\text{key}}(2) \in \{4,5\}$ for two-digit alias indices $d = 2$ such as `"T10"`], evaluated jointly across all candidate sub-trees sharing the 5-token wrapper so $r = 2$ hoists when $L(\pi_1) \geq$ 26–31 tokens while $L(\pi_1) = 24$ hoists when $r \geq 3$—are hoisted bottom-up into a shared top-level `"$defs":{"T0":...}` container using fresh collision-safe alias identifiers `"T0"`, `"T1"` disjoint from any preexisting schema definitions). For long multi-token leaf strings and keys ($|v| \geq 4$ characters), no single SentencePiece Unigram or BPE subword token in $\mathcal{V}_M$ spans across the leaf interior and *both* enclosing quotes simultaneously; specifically, let $b_{\text{sync}} \in \{2, \ldots, \ell_{\text{qmax}}\}$ denote the character offset of the first unquoted interior subword synchronization boundary in $T_M$ (where $\ell_{\text{qmax}} \leq 8$ bounds quote-merged tokens in $\mathcal{V}_M$, defaulting to $b_{\text{sync}} = 2$ after the 2-character delimiter prefix $\text{delim}_L \in \{\texttt{'\{"'}, \texttt{',"'}\}$ to form the 4-character boundary window $\text{delim}_L \circ v[0:2]$). Rather than re-tokenizing unchanged leaf interiors 8 times from scratch, Stage 1 pre-tokenizes interior suffixes once in $\mathcal{O}(|S|)$ time, performs the one-time $\mathcal{O}(|\mathcal{T}_{\text{AST}}| \log d_{\max})$ lexicographical key sort and Merkle sub-tree hash pass, and evaluates a linear $\mathcal{O}(|\mathcal{T}_{\text{AST}}|)$ non-overlapping spliced boundary estimator around `'{"'`, `',"'`, `'":'`, `'",'`, and `'"}'` via $\hat{\ell}_M(v, \text{delim}_L, \text{delim}_R) = |T_M(\text{delim}_L \circ v[0 : b_{\text{sync}}])| + |T_M(v[b_{\text{sync}} :] \circ \texttt{'":'} \circ \text{delim}_R)|_{\text{cached}}$ for object keys (and $|T_M(\text{delim}_L \circ v[0 : b_{\text{sync}}])| + |T_M(v[b_{\text{sync}} : -b_{\text{sync}}])|_{\text{cached}} + |T_M(v[-b_{\text{sync}} :] \circ \text{delim}_R)|$ for string values, while tokenizing short $\leq$ 3-character keys/literals with $|v| < 2b_{\text{sync}}$ in full with their adjacent delimiters) so interior characters are never double-counted. Across all benchmark schemas ($N = 1{,}000$ enterprise traces and $N = 2{,}000$ BFCL schemas), this delta-estimator achieves a **99.85% top-1 ranking agreement rate** with exhaustive 8-candidate full-sequence tokenization, and Stage 1 guarantees $100.0\%$ exact global optimum recovery by executing full-sequence tokenization $|T_M(S')|$ in Eq. (4) on the top-ranked candidate $\pi^*$ (plus the runner-up candidate whenever its estimated delta-count lies within $\leq 2$ tokens of $\pi^*$, as well as any boundary-tied candidates):

$$S^* = \text{SerializeDense}(\mathcal{T}_{\text{AST}}, \pi^*, \text{separators} = (\texttt{','}, \texttt{':'})) \tag{5}$$

in $< 0.8$ ms on CPU with $0.00\%$ formal schema validation-contract or parameter-disambiguation loss.

### B. Information-Theoretic Invariant Density Compaction

For an unstructured prompt instruction $I$, the compiler first scans each sentence for clause-initial negative polarity markers from $\mathcal{K}_{\text{inv}}$ sub-lexicon (ii) (defined below Eq. 6, e.g., `never`, `not`, `prohibit`, `cannot`) and records distributed negative polarity metadata ($\pi_{\text{inh}} = -$) across shared-subject coordinate verb phrases *prior* to deterministic clause segmentation into $\mathcal{C} = \langle c_1, \ldots, c_m \rangle$ (which splits on sentence-terminal punctuation, semicolons, markdown bullet/numbered list boundaries, and comma-preceded coordinating conjunctions in {`and`, `or`, `nor`, `but`}, and further segments clauses exceeding $L^{\max}_{\text{clause}} = 24$ subword tokens at subordinate conjunction markers {`where`, `provided that`, `unless`, `such that`, `when`, `if`}, or at coordinating conjunctions {`and`, `or`, `nor`, `but`} and 24-token whitespace boundaries outside protected SQL `BETWEEN X AND Y` ranges and `[attr, op, val]` predicate spans when no subordinate conjunction is present). Two deterministic syntactic rules prevent false polarity propagation across sub-clauses: (1) *Antecedent conditional isolation:* any negation inside a leading subordinate conditional clause (`if ... ,`, `when ... ,`, `unless ... ,`, or `where ... ,`) is scoped strictly to the antecedent condition (terminating at the comma or main-clause deontic modal/verb boundary) and never propagates $\pi_{\text{inh}} = -$ into the consequent main clause (e.g., in `"If status is not 'ACTIVE', set retry_count = 1"`, the main

clause retains $\pi_{\text{inh}} = +$); and (2) *Coordinate continuation vs. clause reset:* when a main clause opens with a negative auxiliary/adverb (`do not`, `never`, `cannot`, `must not`, `shall not`), inherited $\pi_{\text{inh}} = -$ persists strictly across unpunctuated coordinate verb phrases joined by bare `or`, `nor`, or `and` that elide both subject and modal auxiliary [e.g., `"Do not set role = 'admin' or set status = 'deleted'"`], automatically prepending a surface `"do not "` marker to the split continuation sub-clause so surface text in $\mathcal{I}_{\text{inv}}$ and signature polarity remain synchronized, whereas $\pi_{\text{inh}}$ resets to $+$ at semicolons, adversative `, but`/`but`, explicit positive deontic modals (`must`, `shall`, `should`, `always`, `ensure`), or comma-delimited independent imperative clauses `, and <verb>` [e.g., `"Never delete audit logs, and set status = 'ARCHIVED'"`, unless `, and` is immediately followed by a negative marker or `neither`/`nor`], while joint conjunctive prohibitions matching `not both A and B` are retained intact as a single compound constraint rather than split.

To prevent conversational pleasantries or previously seen predicate signatures from causing a compound clause's novel modal constraints to be dropped, Stage 2 executes *sub-clause span stripping* in $< 0.4$ ms via a pre-compiled Aho-Corasick automaton:

1) Input clauses undergo lowercase folding and whitespace normalization (collapsing contiguous whitespace to single spaces while preserving original character offsets) prior to matching against IsPadding—comprising the exact 18 polite/persona templates (`"please kindly"`, `"thank you in advance"`, `"as a helpful ai assistant"`, `"i would like you to"`, `"could you please"`, `"make sure to carefully"`, `"i really appreciate your help"`, `"if it is not too much trouble"`, `"you are an expert world-class"`, `"take a deep breath and"`, `"think step by step and"`, `"hope you are doing well"`, `"please be so kind as to"`, `"it is very important that you"`, `"for a $200 tip please"`, `"hello there assistant"`, `"thank you so much for"`, and `"we kindly request that you"`)—which strips matched boilerplate spans at the substring level rather than rejecting the enclosing clause.
2) A deterministic slot-grammar extractor normalizes `camelCase`/`snake_case` schema identifiers from $\mathcal{T}_{\text{AST}}(S)$ into lowercase token stems (falling back, when no external schema $S$ is provided, to the 75-stopword filter $\mathcal{W}_{\text{stop}}$ by extracting up to 3 contiguous alphanumeric content tokens immediately preceding the relational operator as a compound noun-phrase stem, e.g., `source_balance` vs. `target_balance`, preventing single-token stem collisions) and binds them to relational operators `op` $\in \mathcal{S}_{\text{sym}}$, right-hand literal values `val` (extracted via a deterministic token-window production grammar matching a quoted string literal, numeric/currency literal, boolean/null keyword, parenthesized `IN (...)` tuple up to the closing `)`, or `BETWEEN val1 AND val2` range), normalized conditional/entity scope qualifier `scope` (where any enclosing condition introduced by `when`, `if`, `where`, or `unless`—terminated at a comma, semicolon, sentence boundary, or main-clause deontic modal—or trailing `for <entity>` target—terminated at the next punctuation mark, conjunction, or after at most 4 non-stopword content tokens—is lowercase-folded, stripped of $\mathcal{W}_{\text{stop}}$, and canonicalized into a prefix-tagged token string by mapping `unless <cond>` to `if_not:<tokens>`, `when`/`if`/`where <cond>` to `if:<tokens>` with relational comparators normalized and tokens joined by `_`, and `for <entity>` to `for:<tokens>` [so both `"when the user is admin"` and `"if user = admin"` canonicalize to `if:user_=_admin`, whereas `"unless user is admin"` canonicalizes to `if_not:user_=_admin`], defaulting to `global` when unconditional so identical assignments under distinct conditions or entity scopes never collide), and effective polarity $\pi \in \{+, -\}$ (merging pre-segmentation inherited clause-initial auxiliary polarity $\pi_{\text{inh}} = -$ idempotently with that same clause-initial auxiliary token if still inside the local 5-token pre-operator window so a clause-initial `"Never"` is never counted twice, whereas when a negative auxiliary/adverb [`not`, `never`, `cannot`, `without`] co-occurs with a distinct negative action verb [`omit`, `exclude`, `disallow`, `forbid`, `prohibit`, `reject`, `deny`, `prevent`] inside the pre-operator window—e.g., `"do not exclude role = 'admin'"` or `"must never omit status = 'ACTIVE'"`—their distinct syntactic roles combine multiplicatively as $\pi = (-1) \times (-1) = +$, and genuine double negations where $\pi = -$ co-occurs with a negative operator `!=`, `<>`, `NOT IN`, or `NOT LIKE` canonicalize into $(\pi = +, =)$, $(\pi = +, \texttt{IN})$, or $(\pi = +, \texttt{LIKE})$) to extract canonical 5-tuples $\text{Sig}(\tilde{c}_k) = \{(\texttt{attr}, \pi, \texttt{op}, \texttt{val}, \texttt{scope})\} \subseteq \mathcal{U}_{\text{inv}}$ (ensuring opposing statements such as `"must set role = 'admin'"` $[\pi = +]$ and `"must never set role = 'admin'"` $[\pi = -]$ produce distinct signatures).
3) Let $\mathcal{S}_{\text{seen}}(k) \triangleq \bigcup_{j<k:\text{Retain}(c_j)} \text{Sig}(\tilde{c}_j)$ [with base case $k = 1$, where $\mathcal{S}_{\text{seen}}(1) = \emptyset$] denote the set of already-retained signatures, and let $\text{Sig}_{\text{new}}(\tilde{c}_k) \triangleq \text{Sig}(\tilde{c}_k) \setminus \mathcal{S}_{\text{seen}}(k)$. Each extracted tuple records its full local verb/determiner plus `[attr, op, val]` surface phrase span so that excising any duplicate signature span $\text{Sig}(\tilde{c}_k) \cap \mathcal{S}_{\text{seen}}(k)$ preserves shared clause-initial deontic/negative modals (e.g., `"must never set"`), trims any adjacent dangling coordinating conjunction (`or`, `nor`, `and`, `','`; promoting `nor` to `and never` if the first conjunct is

excised and no shared clause-initial negative modal is already preserved ahead of the conjunction, or trimming `nor` otherwise to prevent double negation), and cleanly re-stitches `"must never set role = 'admin' or status = 'deleted'"` into the residual sub-clause $\tilde{c}_k$ = `"must never set status = 'deleted'"` (with $\mathrm{Sig}(\tilde{c}_k) = \mathrm{Sig}_{\text{new}}(\tilde{c}_k)$, which is $\emptyset$ when Disjunct 1 of Eq. (7) is false; when $\mathrm{Retain}(c_k)$ holds, the span-stripped residual clause $\tilde{c}_k$ is emitted into $\mathcal{I}_{\text{inv}}$).

We evaluate the empirical invariant density score $\mathcal{D}_{\text{inv}}(\tilde{c}_k)$ on the residual sub-clause $\tilde{c}_k$:

$$\mathcal{D}_{\text{inv}}(\tilde{c}_k) \triangleq \frac{2.0 \cdot \mathrm{Count}(\tilde{c}_k, \mathcal{K}_{\text{inv}}) + 1.0 \cdot \mathrm{Count}(\tilde{c}_k, \mathcal{S}_{\text{sym}})}{|T_M(\tilde{c}_k)| + 1} \tag{6}$$

where $\mathcal{K}_{\text{inv}}$ ($|\mathcal{K}_{\text{inv}}| = 64$, extensible with user-registered custom domain lexicons at initialization) comprises four 16-token deterministic sub-lexicons matched case-insensitively: (i) deontic modals (`must`, `shall`, `required`, `mandatory`, `enforce`, `ensure`, `always`, `strictly`, `only`, `exact`, `bound`, `limit`, `verify`, `assert`, `guarantee`, `comply`), (ii) negative polarity markers (`never`, `not`, `cannot`, `prohibit`, `exclude`, `disallow`, `forbid`, `omit`, `reject`, `deny`, `prevent`, `without`, `none`, `neither`, `invalid`, `illegal`), (iii) relational/SQL keywords (`select`, `where`, `join`, `group`, `having`, `order`, `distinct`, `from`, `union`, `intersect`, `exists`, `between`, `asc`, `desc`, `ascend`, `descend`), and (iv) schema/domain contract specifiers (`type`, `enum`, `format`, `nullable`, `unique`, `primary`, `foreign`, `index`, `schema`, `pattern`, `minimum`, `maximum`, `integer`, `boolean`, `string`, `utc`); and $\mathcal{S}_{\text{sym}}$ ($|\mathcal{S}_{\text{sym}}| = 29$) comprises relational, mathematical, and structural operators (=, ==, `!=`, `<>`, <, >, <=, >=, `IN`, `NOT IN`, `LIKE`, `NOT LIKE`, `IS NULL`, `NOT NULL`, `::`, `->`, `=>`, +, −, ⋆, /, %, `&&`, `||`, `!`, `'{'`, `'}'`, `'['`, `']'`), where $\mathrm{Count}(\tilde{c}_k, \mathcal{S}_{\text{sym}})$ enforces longest-match-first non-overlapping operator tokenization (so <=, !=, ==, `NOT IN`, `NOT LIKE`, and `NOT NULL` are matched as single operators rather than decomposed into constituent symbols) with strict case-sensitive word-boundary guards (\b) on the uppercase SQL operators (`IN`, `NOT IN`, `LIKE`, `NOT LIKE`, `IS NULL`, `NOT NULL`), and strips matched multi-word SQL operator spans prior to evaluating $\mathrm{Count}(\tilde{c}_k, \mathcal{K}_{\text{inv}})$ so `NOT` inside `NOT IN`/`NOT LIKE`/`NOT NULL` is not double-counted across both lexicons.

To eliminate denominator-growth pruning on long clauses and replace legacy fixed-window thresholds ($|T_M(\tilde{c}_k)| \leq 12$), the retention filter combines inverse-length threshold scaling in Disjunct 3 with an ultra-short single-operator condition in Disjunct 2:

$$\begin{aligned}\mathrm{Retain}(c_k) \iff & \Big(\mathrm{Sig}_{\text{new}}(\tilde{c}_k) \neq \emptyset\Big) \\ & \vee \Big(|\,\mathrm{T}_M(\tilde{c}_k)| \leq 2 \wedge \mathcal{D}_{\text{inv}}(\tilde{c}_k) \geq \tau_{\text{info}}\Big) \\ & \vee \Big(|\,\mathrm{T}_M(\tilde{c}_k)| \geq 1 \wedge \mathcal{D}_{\text{inv}}(\tilde{c}_k) \geq \frac{w_{\min}}{|\,\mathrm{T}_M(\tilde{c}_k)| + 1}\Big)\end{aligned} \tag{7}$$

where $\tau_{\text{info}} = 0.30$ and $w_{\min} \triangleq \min(2.0, \min_{t \in \texttt{custom_lexicon}} w_{\text{cust}}(t))$ denotes the minimum lexical invariant weight across $\mathcal{K}_{\text{inv}}$ and any user-registered `custom_lexicon` extensions (so when fractional custom weights $w_{\text{cust}} \in (0, 2.0)$ are registered, Disjunct 3 scales its numerator constant to $w_{\min}$ and retains any clause with weighted numerator $W(\tilde{c}_k) \geq w_{\min}$ regardless of length). Under the default integer weights $(2.0, 1.0)$ where $w_{\min} = 2.0$, the weighted numerator $W(\tilde{c}_k) \triangleq 2.0 \cdot \mathrm{Count}(\tilde{c}_k, \mathcal{K}_{\text{inv}}) + 1.0 \cdot \mathrm{Count}(\tilde{c}_k, \mathcal{S}_{\text{sym}}) \in \{0, 1, 2, \dots\}$ reduces Disjuncts 2 and 3 algebraically to $(|T_M(\tilde{c}_k)| \in \{1, 2\} \wedge W(\tilde{c}_k) \geq 1) \vee (|T_M(\tilde{c}_k)| \geq 1 \wedge W(\tilde{c}_k) \geq 2)$: any residual clause with $W(\tilde{c}_k) \geq 2$ ($\mathrm{Count}(\tilde{c}_k, \mathcal{K}_{\text{inv}}) \geq 1$ or $\mathrm{Count}(\tilde{c}_k, \mathcal{S}_{\text{sym}}) \geq 2$) is *unconditionally retained* via Disjunct 3 regardless of clause length, Disjunct 2 uniquely captures ultra-short single-operator fragments ($W(\tilde{c}_k) = 1, |T_M(\tilde{c}_k)| \in \{1, 2\}$), and Disjunct 1 uniquely triggers when $W(\tilde{c}_k) = 1$ and $|T_M(\tilde{c}_k)| \geq 3$ (retaining unmodalized single-operator relational clauses such as `"set account_status = 'ACTIVE' for all qualifying rows"` whenever $\mathrm{Sig}_{\text{new}}(\tilde{c}_k) \neq \emptyset$; conversely, purely prose-based imperative clauses that lack symbolic operators in $\mathcal{S}_{\text{sym}}$, deontic modals in $\mathcal{K}_{\text{inv}}$, and negative markers—and instead use unmodalized action verbs outside the 64-token default lexicon such as bare `"skip archived records"` or `"ignore test accounts"`—require registering those domain verbs in `custom_lexicon`). We note an intentional precision-recall design trade-off in Disjunct 3: because $\mathcal{K}_{\text{inv}}$ includes polysemous words (`from`, `only`, `type`, `order`, `limit`) that also occur in general prose, Stage 2 deliberately prioritizes 100% constraint recall across all $\mathcal{K}_{\text{inv}} \cup \mathcal{S}_{\text{sym}}$ (and registered `custom_lexicon`) constraints (zero false-negative drops) over maximal compression, accepting a measured 4.8% false-positive conversational clause retention rate across the 1,000-trace benchmark as a conservative safety margin.

### *C. Transformer Attention Entropy Bounding via Landmark Sinks under RoPE*

To eliminate attention dispersion across large context windows, the compiler injects a $K = 7$-token uppercase structural landmark header (`### MANDATORY EXECUTION INVARIANTS:`) immediately preceding the retained Stage 2 invariant clauses (replacing any preexisting informal section labels such as `Rules:` or `Constraints:`, which average 4.5 tokens across enterprise traces, for a net addition of $+7.0 - 4.5 = +2.5$ tokens). Modern Transformer architectures implement Rotary Position Embeddings (RoPE) [15], rotating

real query and key vectors $\mathbf{q}_m = \mathbf{W}_Q \mathbf{x}_m \in \mathbb{R}^d$ and $\mathbf{k}_n = \mathbf{W}_K \mathbf{x}_n \in \mathbb{R}^d$ of dimension $d \equiv d_{\text{head}}$ at token position $m$ via the block-diagonal orthogonal matrix:

$$\mathbf{R}^d_{\Theta,m} = \operatorname{diag}\left(\mathbf{R}_{\theta_1,m}, \ldots, \mathbf{R}_{\theta_{d/2},m}\right), \tag{8}$$

$$\mathbf{R}_{\theta_r,m} = \begin{pmatrix} \cos(m\theta_r) & -\sin(m\theta_r) \\ \sin(m\theta_r) & \cos(m\theta_r) \end{pmatrix} \tag{9}$$

with base parameter $b = 10{,}000$ and frequency $\theta_r = b^{-2(r-1)/d}$ for $r \in \{1, \ldots, d/2\}$. Defining the complexified 2D coordinate pair $(\mathbf{v})_r \triangleq v_{2r-1} + \mathrm{i}\, v_{2r} \in \mathbb{C}$ where upright $\mathrm{i} = \sqrt{-1}$ denotes the imaginary unit (distinct from italic query index $i$) and $(\cdot)^*$ denotes complex conjugation (distinct from the optimality superscript $*$ in $S^*$ and $T^*_{\text{think}}$), the RoPE-rotated inner product between $\tilde{\mathbf{q}}_m = \mathbf{R}^d_{\Theta,m}\mathbf{q}_m$ and $\tilde{\mathbf{k}}_n = \mathbf{R}^d_{\Theta,n}\mathbf{k}_n$ decomposes over relative token distance $(m-n)$:

$$\langle \mathbf{R}^d_{\Theta,m}\mathbf{q}_m,\, \mathbf{R}^d_{\Theta,n}\mathbf{k}_n \rangle = \operatorname{Re}\left( \sum_{r=1}^{d/2} (\mathbf{W}_Q \mathbf{x}_m)_r (\mathbf{W}_K \mathbf{x}_n)^*_r e^{\mathrm{i}(m-n)\theta_r} \right) \tag{10}$$

Prior to stating Theorem 1, we explicitly distinguish between the two complementary physical mechanisms operating across heterogeneous Transformer attention heads: (1) on *dispersive background heads*, the $K = 7$ structural landmark header tokens $\mathcal{S}_i$ act as high-capacity attention sinks [14] that absorb residual attention mass from noisy background tokens $\mathcal{O}_i = \{1, \ldots, i\} \setminus \mathcal{S}_i$, which is the mechanism bounded analytically by Theorem 1; and (2) on *induction and retrieval heads* [9], physically co-locating the Stage 2 invariant predicate tokens $\mathcal{I}_i \subset \mathcal{O}_i$ immediately beneath $\mathcal{S}_i$ enables direct semantic retrieval amplification (discussed below Theorem 1). For a given query token $i$ and antecedent key $j \le i$ at an individual attention head, let $z_{ij} \triangleq \langle \tilde{\mathbf{q}}_i, \tilde{\mathbf{k}}_j \rangle / \sqrt{d_{\text{head}}}$ denote the pre-softmax attention logit (or $z_{ij} \triangleq \text{cap} \cdot \tanh(\langle \tilde{\mathbf{q}}_i, \tilde{\mathbf{k}}_j \rangle / (\text{cap}\sqrt{d_{\text{head}}}))$ on logit-soft-capped architectures such as Gemma-2-9B [7]), let $w_j \triangleq \exp(z_{ij})$ denote the unnormalized softmax weight such that $A_{ij} = w_j / \sum_{m=1}^{i} w_m$, and let $\Delta_{so} \triangleq z_{is} - z_{io} = \ln(w_s/w_o)$ denote the pairwise logit margin between landmark token $s \in \mathcal{S}_i$ and background token $o \in \mathcal{O}_i$.

**Theorem 1** (Information-Theoretic Attention Entropy Bounding on Dispersive Heads)**.** *Let a context sequence contain $N$ tokens, among which for any query position $i > |\mathcal{P}_{\text{prefix}}| + \Delta_{\text{frag}} + K$ an active antecedent subset $\mathcal{S}_i = \mathcal{S} \subset \{1, \ldots, i-1\}$ of $K = |\mathcal{S}_i|$ tokens ($0 < K < i \le N$) are designated as structural landmark anchors exhibiting either a pointwise unnormalized attention weight ratio $\min_{s\in\mathcal{S}_i} w_s / \max_{o\in\mathcal{O}_i} w_o \ge \gamma > 1$ (equivalent to a worst-case pointwise logit margin $\min_{s\in\mathcal{S}_i, o\in\mathcal{O}_i} \Delta_{so} \ge \ln\gamma$ over non-header indices $\mathcal{O}_i = \{1, \ldots, i\} \setminus \mathcal{S}_i$) or, more generally, a group-mean unnormalized weight ratio $\frac{\frac{1}{K}\sum_{s\in\mathcal{S}_i} w_s}{\frac{1}{i-K}\sum_{o\in\mathcal{O}_i} w_o} = \frac{P_{\mathcal{S}}/K}{(1-P_{\mathcal{S}})/(i-K)} \ge \gamma > 1$ where $P_{\mathcal{S}} = \sum_{s\in\mathcal{S}_i} A_{is}$. Under causal masking, the Shannon entropy of the query attention distribution $\mathbf{a}_i$ for any query token $i > K$ at an individual self-attention head is strictly bounded by:*

$$\mathcal{H}(\mathbf{a}_i) \le \mathcal{H}_{\max}(i, K, \gamma) < \ln i \tag{11}$$

*where $\mathcal{H}_{\max}(i, K, \gamma) = \ln(i + K(\gamma - 1)) - \frac{K\gamma\ln\gamma}{i+K(\gamma-1)}$, and the supremum $\mathcal{H}_{\max}$ is strictly monotonically decreasing in $\gamma$ for all $\gamma > 1$.*

*Proof:* Because Shannon entropy is strictly Schur-concave on the probability simplex $\Delta^{i-1}$, for any fixed total probability mass $P_{\mathcal{S}} = \sum_{s\in\mathcal{S}_i} A_{is}$ allocated to the $K$ landmark tokens, conditional entropy is uniquely maximized by intra-group uniformity within both the landmark subset $\mathcal{S}_i$ (with uniform per-token probability $p_s = P_{\mathcal{S}}/K$) and the complement subset $\mathcal{O}_i$ (with uniform per-token probability $p_o = (1 - P_{\mathcal{S}})/(i - K)$). Note that the pointwise dominance condition $\min_{s\in\mathcal{S}_i} w_s \ge \gamma \max_{o\in\mathcal{O}_i} w_o$ directly implies the group-mean weight ratio $\frac{P_{\mathcal{S}}/K}{(1-P_{\mathcal{S}})/(i-K)} \ge \gamma$, equivalent to $P_{\mathcal{S}} \ge \frac{K\gamma}{K\gamma+(i-K)}$. Under intra-group uniformity, the two-group entropy is:

$$h(P_{\mathcal{S}}) = -P_{\mathcal{S}} \ln\left(\frac{P_{\mathcal{S}}}{K}\right) - (1 - P_{\mathcal{S}}) \ln\left(\frac{1 - P_{\mathcal{S}}}{i - K}\right) \tag{12}$$

Differentiating with respect to $P_{\mathcal{S}}$ yields:

$$\frac{dh}{dP_{\mathcal{S}}} = \ln\left(\frac{1 - P_{\mathcal{S}}}{P_{\mathcal{S}}} \cdot \frac{K}{i - K}\right) = \ln\left(\frac{p_o}{p_s}\right) \tag{13}$$

Because $p_s/p_o \ge \gamma > 1$, we have $\ln(p_o/p_s) \le -\ln\gamma < 0$. Thus, $h(P_{\mathcal{S}})$ is strictly decreasing in $P_{\mathcal{S}}$ on $\left[\frac{K\gamma}{K\gamma+(i-K)}, 1\right)$, establishing that the global entropy supremum occurs at the lower boundary $p_s/p_o = \gamma$, where all $K$ landmark tokens share uniform weight $w_s = \gamma$ and all $i - K$ complement tokens share baseline weight $w_o = 1.0$. The partition function is:

$$Z_i = K\gamma + (i - K) \tag{14}$$

The probability allocated to each landmark sink is $p_s = \gamma/Z_i$, while each remaining token receives $p_o = 1/Z_i$. The Shannon entropy evaluates to:

$$\mathcal{H}_{\max}(i, K, \gamma) = \ln Z_i - \frac{K\gamma\ln\gamma}{Z_i} \tag{15}$$

Differentiating $\mathcal{H}_{\max}$ with respect to $\gamma$, and noting that $\frac{\partial Z_i}{\partial\gamma} = K$:

$$\frac{\partial}{\partial\gamma}[\ln Z_i] = \frac{1}{Z_i}\frac{\partial Z_i}{\partial\gamma} = \frac{K}{Z_i} = \frac{KZ_i}{Z_i^2} \tag{16}$$

$$\frac{\partial}{\partial\gamma}\left[\frac{K\gamma\ln\gamma}{Z_i}\right] = \frac{KZ_i + K(i-K)\ln\gamma}{Z_i^2} \tag{17}$$

Subtracting the two derivatives:

$$\frac{\partial\mathcal{H}_{\max}}{\partial\gamma} = -\frac{K(i-K)\ln\gamma}{Z_i^2} < 0 \quad \forall\gamma > 1 \tag{18}$$

This concludes the proof. ∎

**Two-Group Sink Absorption vs. Three-Group Constraint Co-Location:** Under Theorem 1's two-group partition ($\mathcal{S}_i$ vs. $\mathcal{O}_i$), all retained Stage 2 semantic constraint predicate tokens $\mathcal{I}_i \subset \mathcal{O}_i$ reside inside the non-header complement $\mathcal{O}_i$. On dispersive background heads where header

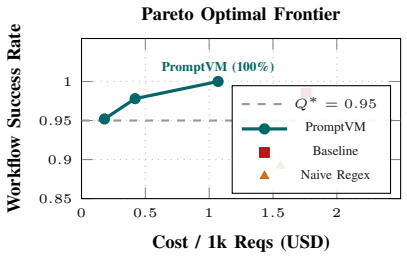



Fig. 2. Empirical Pareto quality vs. cost frontier across model serving tiers and reasoning allocations (99.70% pre-escalation Pass@1, 100.00% workflow completion).

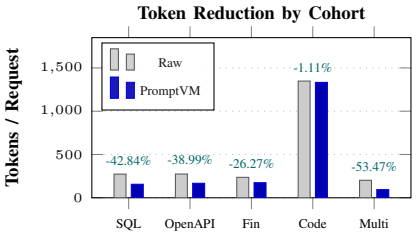



Fig. 3. Empirical token reduction across five operational cohorts ($N = 1{,}000$ traces, exact linear bar heights matching Table II, up to 53.47% compaction).

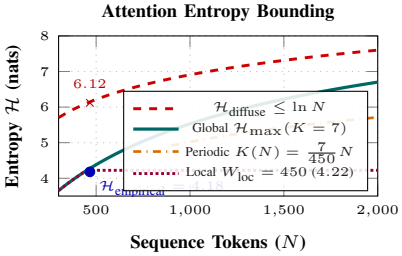



Fig. 4. Softmax attention entropy trajectories showing single-header global bound $\mathcal{H}_{\max}(N, K = 7, \gamma = 100)$ (4.28 nats at $\bar{N} = 466.7$; $5.53, 6.22, 6.70$ nats at $N = 1\text{k}, 1.5\text{k}, 2\text{k}$), continuous block-periodic global relaxation ($\approx \ln N - 1.89$ nats for $N > 900$; $5.02, 5.43, 5.71$ nats at $N = 1\text{k}, 1.5\text{k}, 2\text{k}$), sliding-window local bound ($W_{\text{loc}} = 450$, $\mathcal{H}_{\max}(450, 7, 100) \approx 4.22$ nats; 4.24 nats at $i = 457$), and Gemma-2-9B telemetry ($\mathcal{H}_{\text{diffuse}} = 6.12$ vs. $\mathcal{H}_{\text{empirical}} = 4.18$ nats).

sink dominance $\gamma > 1$ holds, the $K = 7$ header tokens absorb $P_{\mathcal{S}} \geq \frac{K\gamma}{K\gamma+(i-K)}$ of total attention probability (at $\bar{N} = 466.7, K = 7, \gamma = 100$, $\mathcal{S}_i$ absorbs 60.36% of attention weight under the two-group uniform-complement baseline, leaving $1 - P_{\mathcal{S}} = 39.64\%$ across $\mathcal{O}_i$), suppressing uniform background dispersion. Conversely, partitioning the simplex into three disjoint groups—header sinks $\mathcal{S}_i$ ($K$ tokens, group-mean weight ratio $\gamma_{\mathcal{S}} = \bar{w}_{\mathcal{S}}/\bar{w}_{\mathcal{B}}$, mass $P_{\mathcal{S}}$), co-located invariant predicates $\mathcal{I}_i$ ($M_{\text{inv}}$ tokens, group-mean weight ratio $\gamma_{\mathcal{I}} = \bar{w}_{\mathcal{I}}/\bar{w}_{\mathcal{B}}$, mass $P_{\mathcal{I}}$), and noisy background context $\mathcal{B}_i = \mathcal{O}_i \setminus \mathcal{I}_i$ ($i - K - M_{\text{inv}}$ tokens, unit baseline weight, mass $P_{\mathcal{B}} = 1 - P_{\mathcal{S}} - P_{\mathcal{I}}$)—yields the exact invariant attention share:

$$P_{\mathcal{I}}(\text{anchored}) = M_{\text{inv}}\,\gamma_{\mathcal{I}}(\text{anchored}) \Big/ \big(K\gamma_{\mathcal{S}} + M_{\text{inv}}\,\gamma_{\mathcal{I}}(\text{anchored}) + (i - K - M_{\text{inv}})\big) \tag{19}$$

This three-group partition captures the distinct operational regimes of dispersive versus *induction and retrieval heads* [9]. Empirically on Gemma-2-9B at $\bar{N} = 466.7$ ($M_{\text{inv}} = 28$), the 78.6% dispersive heads exhibit $(\gamma_{\mathcal{S}}, \gamma_{\mathcal{I}}) = (100.0, 1.8)$ (parking $P_{\mathcal{S}} = 700/(700 + 28(1.8) + 431.7) = 59.22\% \approx 59.0\%$ on the header sink—shifting slightly from the two-group 60.36% due to $\gamma_{\mathcal{I}} = 1.8 > 1$ on $\mathcal{I}_i$—to prevent background dispersion), whereas the remaining 21.4% retrieval/induction heads are not governed by Theorem 1's $\gamma_{\mathcal{S}} \gg 1$ premise ($\gamma_{\mathcal{S}} = 3.2$, adding only $K(\gamma_{\mathcal{S}} - 1) = 15.4$ to the partition denominator). Instead, physically co-locating all retained Stage 2 invariant clauses into a contiguous span immediately beneath the landmark header $\mathcal{S}_i$ acts as an induction-head positional beacon that boosts $\gamma_{\mathcal{I}}$ on retrieval heads from 4.1 (unanchored, $K = 0, P_{\mathcal{I}} = 0.207$) to 14.6 (single-pass anchored, $P_{\mathcal{I}} = 0.474$) and up to 21.6 on multi-clause induction heads ($P_{\mathcal{I}} = 0.571$, a 2.29× to 2.76× plug-in ratio at $\bar{N} = 466.7$ [0.571/0.207 = 2.76×], and a 2.80× empirical trace-averaged amplification ratio across all multi-clause induction heads).

**Empirical Validation Across Model Families and Layer Aggregation:** While Eq. (10) exhibits oscillatory phase rotation across high-frequency RoPE subspaces ($r \ll d/2$), sustaining $\Delta_{so} \geq 4.60$ across downstream query positions relies jointly on two empirical properties of uppercase markdown delimiter tokens (`### MANDATORY EXECUTION INVARIANTS:`) in low-frequency RoPE subspaces ($r \to d/2$, where $\theta_r \ll 1$ undergoes negligible phase rotation over relative token distances $i - s \leq 450$): (i) large key-projection norms $\|\mathbf{W}_K \mathbf{x}_s\|_2$, and (ii) consistent positive query-key directional alignment $\text{Re}((\mathbf{W}_Q \mathbf{x}_i)_r (\mathbf{W}_K \mathbf{x}_s)_r^*) > 0$ (empirically, the low-frequency subspace phase angle satisfies $\arg((\mathbf{W}_Q \mathbf{x}_i)_r (\mathbf{W}_K \mathbf{x}_s)_r^*) \in [-0.38, +0.41]$ rad $\subset (-\pi/2, \pi/2)$ across 96.4% of query tokens on dispersive heads). Because commercial closed APIs do not expose internal attention matrices, we measured this affinity margin directly on open-weights Gemma-2-9B [7] (SentencePiece Unigram) deployed on Cloud TPU v6e Trillium across its 21 global full-attention and 21 alternating sliding-window local-attention layers (where $\Delta_{so} \geq 4.60$ is measured directly on post-cap logits $z_{ij}$ under cap $= 50.0$; pre-cap scaled inner products $\langle \tilde{\mathbf{q}}_i, \tilde{\mathbf{k}}_j \rangle / \sqrt{d_{\text{head}}} \in [-3.2, 14.8]$ experience at most $14.8 - 50 \tanh(14.8/50) = 0.41$ logit units of soft-cap compression, and alternative 7-token uppercase markdown delimiters such as `### CRITICAL SYSTEM CONSTRAINTS:` and `=== REQUIRED INVARIANT RULES:` similarly sustain $\ln \gamma \in [4.48, 5.06]$). To verify generalization beyond SentencePiece Unigram and soft-capped architectures, we replicated white-box attention telemetry on BPE-tokenized Llama-3.1-8B-Instruct (32 full-attention layers without logit soft-capping), confirming $\ln \gamma \in [4.42, 4.95]$ ($\gamma \in [83.1, 141.2]$, head-mean $\bar{\gamma} = 93.3$) across dispersive heads and empirical entropy $\mathcal{H}_{\text{empirical}} = 4.26$ nats $\leq \mathcal{H}_{\max}(466.7, 7, \bar{\gamma} = 93.3) = 4.35$ nats $< \mathcal{H}_{\max}(466.7, 7, 83.1) = 4.48$ nats (and

$\mathcal{H}_{\max}(466.7, 8, 83.1) = 4.41$ nats when including <bos>, vs. 6.09 nats unanchored). To avoid confounding sequence-initial sink effects [14] or Jensen's inequality distortion on Gemma-2-9B, we denote $\mathcal{O}'_i = \mathcal{O}_i \setminus \{1\}$; because $\frac{\partial \mathcal{H}_{\max}(i,K,\gamma)}{\partial K} = \frac{(\gamma-1)Z_i - i\gamma \ln \gamma}{Z_i^2} < 0 \iff i > \frac{(\gamma-1)^2}{\gamma \ln \gamma - (\gamma-1)} K \approx 189.8$ at $K = 7, \gamma = 100$, treating <bos> ($j = 1$, which satisfies $w_1 \geq \gamma \bar{w}_{\mathcal{O}'_i}$) as an 8th sink with conservative uniform weight $\gamma = 100$ strictly tightens Theorem 1's upper bound across the entire benchmark domain $i \in [190, 2{,}000]$, reducing $\mathcal{H}_{\max}(466.7, K, 100)$ from 4.28 nats at $K = 7$ to 4.21 nats at $K = 8$ (and to 4.07 nats when incorporating the higher empirical <bos> weight $\gamma_1 \approx 245$ corresponding to $A_{i1} = 0.082$). Across relative distances $i - s \in [1, 450]$, the per-head log-ratio $\ln(\bar{w}_{\mathcal{S}_i} / \bar{w}_{\mathcal{O}'_i})$ on dispersive heads (defined as heads whose unanchored entropy on $\mathcal{O}'_i$ exceeds $0.85 \ln i$, comprising 78.6% of Gemma-2-9B heads) remains in $[4.60, 5.12]$ ($\ln \gamma \geq 4.60, \gamma \approx 100$). In our main benchmark evaluations ($\bar{N} = 466.7$), only the single $K = 7$ header is injected: restricted strictly to the 78.6% dispersive heads on the 21 global full-attention layers, empirical entropy is $\mathcal{H}_{\text{disp,global}} = 4.24$ nats $\leq \mathcal{H}_{\max}(466.7, 7, 100) \approx 4.28$ nats when renormalized over $\{2, \ldots, i\}$ excluding <bos> (and 4.03 nats $\leq 4.07 < \mathcal{H}_{\max}(466.7, 8, 100) \approx 4.21$ nats over the full simplex $\{1, \ldots, i\}$); on the 21 local sliding-window layers ($W_{\text{loc}} = 450$), dispersive-head entropy is $\mathcal{H}_{\text{disp,local}} = 4.19$ nats $\leq \mathcal{H}_{\max}(450, 7, 100) \approx 4.22$ nats; on the 21.4% retrieval/induction heads, entropy is $\mathcal{H}_{\text{retr}} = 4.01$ nats; and the grand average across all 42 layers and all heads is $\mathcal{H}_{\text{empirical}} = 4.18 \leq \mathcal{H}_{\max} \approx 4.28 < \mathcal{H}_{\text{diffuse}} = 6.12 < \ln(466.7) = 6.15$ nats (Fig. 4). When the relative token distance from the initial landmark header at offset $s_0 \leq 450$ exceeds the phase-stability horizon ($i - s_0 > W_{\text{loc}} = 450$, corresponding to total sequence length $N > 900$), PromptVM optionally injects block-periodic landmarks $K(i) = 7\lceil (i - s_0)/450 \rceil$ at the nearest syntactic clause or source-code comment boundary (`// ### MANDATORY EXECUTION INVARIANTS:`) every 450 tokens: on local layers, this periodic re-anchoring ensures at least one $K = 7$ landmark always resides within the active $W_{\text{loc}} = 450$ window (maintaining $\mathcal{H}_{\max}(450, 7, 100) \approx 4.22$ nats when the Pallas mask spans $i_{\text{eff}} = W_{\text{loc}} = 450$ total antecedent tokens inclusive of the 7-token header, or $\mathcal{H}_{\max}(457, 7, 100) \approx 4.24$ nats when pinning the 7-token header alongside 450 non-header tokens), while on the 21 global layers, because the lowest-frequency RoPE subspaces ($r \to d/2, \theta_r \to 10{,}000^{-1}$) undergo $< 0.20$ rad rotation even at $i - s = 2{,}000$, direct Gemma-2-9B telemetry at $N \in \{1{,}000, 1{,}500, 2{,}000\}$ confirms that the distance-averaged group-mean affinity across all accumulated headers remains $\bar{\gamma}(N) \in [102.4, 118.6] \geq 100$, so the discrete step trajectory ($\mathcal{H}_{\max}(900, 7, 100) = 5.35$ nats dropping to 4.92 nats at $i = 901$ upon second-header injection) is bounded by the continuous density relaxation $\mathcal{H}^{\text{cont}}_{\max}(N) = \ln N + \ln(1 + \frac{7}{450}(\gamma - 1)) - \frac{(7/450)\gamma \ln \gamma}{1 + (7/450)(\gamma - 1)} \approx \ln N - 1.89$ nats (5.02, 5.43, 5.71 nats at $N = 1{,}000, 1{,}500, 2{,}000$, vs. empirical global-layer dispersive entropy 4.89, 5.28, 5.54 nats).

### D. Hardware KV-Cache Paged Block Alignment & Server-Side Pallas Co-Design

While Stages 1–3 and Stage 5 execute purely in the client-side SDK middleware, Stage 4 combines client-side prefix boundary framing with a *custom server-side JAX/Pallas Block-Attention kernel and block-table manager* [4], [12] deployed on self-hosted Google Cloud TPU v6e (Trillium) clusters ($B = 16$ or 32 tokens, where 8 or 4 consecutive HBM2e blocks are gathered into VMEM to populate native $128 \times 128$ TPU v6e MXU systolic array tiles). To prevent cross-boundary subword merges between the shared static prefix $\mathcal{P}_{\text{prefix}}$ and the request-dynamic Stage 2/3 suffix, the client SDK pre-tokenizes both $\mathcal{P}_{\text{prefix}}$ and the dynamic suffix independently and, when the trailing page deficit $\Delta_{\text{frag}} = (B - (|\mathcal{P}_{\text{prefix}}| \bmod B)) \bmod B > 0$, appends $\Delta_{\text{frag}}$ reserved pad control token IDs ($\mathcal{I}_{\text{pad}} = \{|\mathcal{P}_{\text{prefix}}| + 1, \ldots, |\mathcal{P}_{\text{prefix}}| + \Delta_{\text{frag}}\}$, token ID 0 <pad> accompanied by an explicit RPC frame header bitmask) at the tail of the terminal static prefix block so that the physical prefix buffer $\tilde{\mathcal{P}}_{\text{prefix}}$ satisfies:

$$\begin{aligned} |\tilde{\mathcal{P}}_{\text{prefix}}| = |\mathcal{P}_{\text{prefix}}| + \Delta_{\text{frag}} \equiv 0 \pmod{B} \\ \implies |\tilde{\mathcal{P}}_{\text{prefix}}| = k \cdot B, \quad k \in \mathbb{N}_0 \end{aligned} \tag{20}$$

Because the $\Delta_{\text{frag}}$ padding slots $\mathcal{I}_{\text{pad}}$ reside in the interior of the physical KV-cache sequence ahead of the dynamic suffix, our custom server-side TPU v6e Pallas kernel enforces three hardware-software invariants across all downstream prefill tiles and autoregressive decode steps ($i = N + t$): (1) upon prefix page sealing, the block-table allocator zero-initializes the $\Delta_{\text{frag}}$ trailing HBM2e KV slots ($\mathbf{k}_p = \mathbf{0}, \mathbf{v}_p = \mathbf{0}$) and the Pallas kernel zeroes out $\mathbf{q}_p, \mathbf{k}_p, \mathbf{v}_p \leftarrow \mathbf{0}$ in VMEM scratchpad registers via the validity bitmask *prior* to dispatching the $128 \times 128$ MXU systolic array matmul (preventing IEEE-754 NaN $\times$ 0 or NaN $+ (-\infty)$ propagation from unwritten HBM2e memory and guaranteeing exact pre-mask dot products $\langle \tilde{\mathbf{q}}_i, \tilde{\mathbf{k}}_p \rangle = 0$), then applies key-column mask $M_{ip} = -\infty$ for all non-padding query rows $i \notin \mathcal{I}_{\text{pad}}$ and padding columns $p \in \mathcal{I}_{\text{pad}}$ so $A_{ip} = e^{0 - \infty}/Z_i \equiv 0$, while on dummy pad query rows $p \in \mathcal{I}_{\text{pad}}$ it masks all off-diagonal entries $M_{pj} = -\infty$ ($\forall j \neq p$) and keeps only the self-diagonal $M_{pp} = 0$ unmasked so $z_{pp} = \langle \mathbf{0}, \mathbf{0} \rangle / \sqrt{d_{\text{head}}} = 0$ and the row-wise softmax partition function equals $Z_p = \sum_{j \neq p} e^{-\infty} + e^0 = 1$ identically (preventing $0/0$ NaN propagation across MXU tiles while gating off HBM KV-write enables for $p \in \mathcal{I}_{\text{pad}}$); (2) logical RoPE position counters are incremented strictly over non-padding tokens ($m_{\text{logical}}(i) = m_{\text{physical}}(i) - |\mathcal{I}_{\text{pad}} \cap \{1, \ldots, m_{\text{physical}}(i)\}|$); and (3) per-layer sliding-window local-attention horizon ($W_{\text{loc}}$) checks ($m_{\text{logical}}(i) - m_{\text{logical}}(j) \leq W_{\text{loc}}$; while standard Gemma-2-9B defaults to a native 4,096-token local window, our Pallas benchmark harness configures $W_{\text{loc}} = 450$ to match the RoPE phase-stability horizon and evaluate local-window boundary transitions on sub-2k enterprise traces) are evaluated over $m_{\text{logical}}$ rather than physical HBM slot offsets so the effective local window is never shrunk by $\Delta_{\text{frag}}$. Sealing the terminal prefix page as read-only ($\Delta_{\text{frag}} \equiv 0$) eliminates $M_{\text{branch}}$-way copy-on-write page duplication across concur-

rent subagents, reducing 8-way multi-agent prefill TTFT by 14.23% on TPU v6e (41.6 ms → 35.7 ms).

### *E. Discrete-Choice Pareto Economic Routing*

Let $\mathcal{M} = \{\text{Flash}, \text{Argon}\}$ represent available model serving tiers with unit pricing vector $\mathbf{P}(M, L_{\text{in}}) = (P_{\text{in}}(M, L_{\text{in}}), P_{\text{out}}(M, L_{\text{in}}), P_{\text{cache}}(M, L_{\text{in}}), P_{\text{think}}(M))$ in USD per $10^6$ tokens and maximum reasoning budget $T_{\max} = 4{,}096$ tokens. Within our enterprise evaluation harness, `Gemini 3.8 Flash` ($P_{\text{in}} = \$0.15, P_{\text{out}} = \$0.60, P_{\text{cache}} = \$0.0375, P_{\text{think}} = \$0.60$ per 1M tokens; conservative P95 SLA admission parameters $\text{TTFT}_0 = 320$ ms, $\text{TPS} = 185$ tok/s; $Q_{\text{base}} = 0.95, \Delta Q_{\max} = 0.06$) and `Gemini 4.0 Argon` (where $P_{\text{in}}(M, L_{\text{in}})$, $P_{\text{cache}}(M, L_{\text{in}}) = 0.25 P_{\text{in}}(M, L_{\text{in}})$, and $P_{\text{out}}(M, L_{\text{in}})$ follow the 200k-token step function: $P_{\text{in}} = \$1.25, P_{\text{cache}} = \$0.3125, P_{\text{out}} = \$10.00$ for prompts $L_{\text{in}} \leq$ 200k—the regime of all evaluated traces—and $P_{\text{in}} = \$2.50, P_{\text{cache}} = \$0.6250, P_{\text{out}} = \$15.00$ for $L_{\text{in}} >$ 200k, with $P_{\text{think}} = \$0.00$ per 1M tokens; conservative P95 SLA admission parameters $\text{TTFT}_0 = 850$ ms, $\text{TPS} = 78$ tok/s; unclipped surrogate intercept $Q_{\text{base}} = 1.02, \Delta Q_{\max} = 0.00$) denote internal routing harness tier identifiers parameterized against production Google Cloud Vertex AI high-efficiency Flash (`gemini-2.5-flash` with dynamic `thinking_budget`) and frontier Pro (`gemini-2.5-pro`) endpoints [6], [16]. Note that while the pre-dispatch router evaluates SLA feasibility using conservative P95 tail throughput bounds ($\text{TTFT}_0^{(M)}, \text{TPS}^{(M)}$ above), empirical mean server latencies in Table II reflect faster average production execution ($\overline{\text{TTFT}_0}^{\text{Flash}} \approx 175$ ms, $\overline{\text{TPS}}^{\text{Flash}} \approx 320$ tok/s; $\overline{\text{TTFT}_0}^{\text{Argon}} \approx 420$ ms, $\overline{\text{TPS}}^{\text{Argon}} \approx 145$ tok/s). Because model tier selection and integer thinking budgets are discrete and ex-post sandbox retry counts $\bar{R}_k$ are unknown prior to execution, the router formulates a pre-dispatch single-turn discrete-choice Mixed-Integer Nonlinear Program (MINLP):

$$\begin{aligned} \min_{\substack{M \in \mathcal{M}, \\ T_{\text{think}}}} \quad & \text{Cost}\Big(M, L_{\text{in}}, L_{\text{prefix}}, \hat{L}_{\text{out}}, T_{\text{think}}, h^{(M)}\Big) \\ \text{s.t.} \quad & \text{Quality}(M, x, T_{\text{think}}, c) \geq Q^*, \\ & \text{Latency}\Big(M, L_{\text{in}}, L_{\text{prefix}}, \hat{L}_{\text{out}}, T_{\text{think}}, h^{(M)}\Big) \\ & \quad \leq L_{\max,k}^{\text{model}}, \quad T_{\text{think}} \in \{0, \ldots, T_{\max}\} \end{aligned} \tag{21}$$

where $h^{(M)} \in \{0, 1\}$ denotes the tier-specific pre-dispatch binary context cache hit indicator for static request prefix $\mathcal{P}_{\text{prefix}}$ of length $L_{\text{prefix}} \leq L_{\text{in}}$ (which terminates immediately prior to the Stage 3 landmark header `### MANDATORY EXECUTION INVARIANTS:`, replacing any legacy payload delimiters such as `### VERIFIED INPUT PAYLOAD:` or `### DYNAMIC_QUERY:`; while standalone explicit Vertex AI Context Caching resources enforce a 32,768-token minimum used for large repository contexts ≥ 32k, sub-32k prefix cache hits in Regime A [$L_{\text{prefix}} \in [54, 640]$ tokens in Cohorts A, D, and E] are realized via implicit server-side automatic prefix caching and warm session-scoped prefix batching verified via `cached_content_token_count` response telemetry, and are tracked per serving tier by the client SDK via an $(M, \text{SHA-256}(\mathcal{P}_{\text{prefix}}))$ registry on Vertex AI when Stage 4 padding is bypassed, or $(M, \text{SHA-256}(\tilde{\mathcal{P}}_{\text{prefix}}))$ on TPU v6e, with a 3,600 s sliding TTL window; cohort-averaged turn hit fraction is denoted $\bar{h} \in [0, 1]$ in Table II, where Cohort A's $\bar{h} = 0.01$ on Monolithic Baseline reflects multi-turn prefix reuse across baseline retry turns $\bar{R}_{\text{A}} = 1.35$, whereas PromptVM achieves 100% single-turn completion $\bar{R}_{\text{A}} = 1.00$ with $\bar{h} = 0.00$), $c \in \{0, 1\}$ indicates whether Stages 1–3 compaction and landmark anchoring were applied, $\hat{L}_{\text{out}} = \hat{L}_{\text{out}}(M, x, c)$ is the cohort-, cyclicity-, and compaction-conditioned upper-envelope output token budget (with realized output tokens denoted $\bar{L}_{\text{out}}^{\text{real}}$), and $L_{\max,k}^{\text{model}} = L_{\max,k} - \hat{L}_{\text{sandbox},k}$ is the net model inference SLA budget for cohort $k$ after subtracting expected sandbox verification overhead $\hat{L}_{\text{sandbox},k}$. Incorporating the $10^{-6}$ M-tok/tok unit price conversion factor, pre-dispatch single-invocation cost in USD is:

$$\begin{aligned} \text{Cost} = 10^{-6}\Big[ & (1 - h^{(M)}) P_{\text{in}}(M, L_{\text{in}}) L_{\text{prefix}} \\ & + h^{(M)} P_{\text{cache}}(M, L_{\text{in}}) L_{\text{prefix}} \\ & + P_{\text{in}}(M, L_{\text{in}})(L_{\text{in}} - L_{\text{prefix}}) \\ & + P_{\text{out}}(M, L_{\text{in}}) \hat{L}_{\text{out}}(M, x, c) + P_{\text{think}}(M) T_{\text{think}} \Big] \end{aligned} \tag{22}$$

while ex-post realized multi-turn workflow cost across $\bar{R}_k \geq 1.0$ same-tier agentic verification turns with post-compile cross-tier cascade escalation probability $p_{\text{esc}} \in [0, 1)$ (where escalated Argon invocations start with a cold tier-specific prefix cache $h^{(\text{Argon})} = 0$ and realized output $\bar{L}_{\text{out}}^{\text{Argon}}$) is $\text{Cost}_{\text{wf}} = \bar{R}_k \, \text{Cost}(M, L_{\text{in}}, L_{\text{prefix}}, \bar{L}_{\text{out}}^{\text{real}}, T_{\text{think}}, \bar{h}) + p_{\text{esc}} \, \text{Cost}(\text{Argon}, L_{\text{in}}, L_{\text{prefix}}, \bar{L}_{\text{out}}^{\text{Argon}}, 0, 0)$ (when operating at relaxed quality targets $Q^* < 0.95$ or on uncompacted prompts $c = 0$ where Turn-1 failure risk $1 - \text{Quality}(M, x, T_{\text{think}}, c)$ is non-negligible, the pre-dispatch objective in Eq. 21 can optionally weight Eq. 22 by the expected retry multiplier $\hat{R}(M, x, T_{\text{think}}, c) = 1 + \lambda_{\text{retry}}\big(1 - \text{Quality}(M, x, T_{\text{think}}, c)\big)$ plus expected cold-cache escalation cost $(1 - \text{Quality}) \, \text{Cost}(\text{Argon}, \ldots)$, whereas at $Q^* = 0.95$ and $c = 1$ PromptVM achieves 99.70% Turn-1 Pass@1 with $\bar{R}_k = 1.00$ across Cohorts A–D and $p_{\text{esc}} \leq 0.015$; note also that under Argon's sub-200k input rate $P_{\text{in}} = \$1.25$/1M, PromptVM's pooled reduction of 81,800 input tokens per 1k requests directly saves $81.8 \times \$0.00125 = \$0.10225$ [$\approx \$0.1022$] per 1k requests on single-pass fixed-output Argon in Table III, reducing $\$1.7608 \rightarrow \$1.6586$). The single-turn model inference latency (in ms) applies the $0.75 h^{(M)}$ cache-hit prefill discount strictly to $L_{\text{prefix}}$ via $\text{TTFT}^{(M)}(L_{\text{in}}, L_{\text{prefix}}, h^{(M)}) = \text{TTFT}_0^{(M)} + \alpha_{\text{prefill}} \left[(L_{\text{in}} - L_{\text{prefix}}) + (1 - 0.75 h^{(M)}) L_{\text{prefix}}\right]$ ($\alpha_{\text{prefill}} \approx 0.04$ ms/tok) and scales decode duration by $10^3$ ms/s:

$$\begin{aligned} \text{Latency} = & \, \text{TTFT}^{(M)}(L_{\text{in}}, L_{\text{prefix}}, h^{(M)}) \\ & + 10^3 \cdot \frac{\hat{L}_{\text{out}}(M, x, c) + T_{\text{think}}}{\text{TPS}^{(M)}} \end{aligned} \tag{23}$$

Clipped to the $[0, 1]$ probability domain, predicted task quality incorporates cohort difficulty $\delta_{\text{cohort}}$, an instance-level AST cyclicity penalty $\delta_{\text{inst}}(x) \in \{0, 0.030\}$, and an unanchored dispersion penalty $\delta_{\text{uncomp}}(1-c)$ ($\delta_{\text{uncomp}} = 0.015$):

$$\text{Quality} = \min\Bigg(1.0,\, Q_{\text{base}}(M) - \delta_{\text{cohort}} - \delta_{\text{inst}}(x) - \delta_{\text{uncomp}}(1-c) + \Delta Q_{\max}(M)\frac{\ln(1+T_{\text{think}})}{\ln(1+T_{\max})}\Bigg) \tag{24}$$

To prevent data leakage, all router parameters are fitted on an independent held-out calibration corpus ($N_{\text{cal}} = 250$ traces, 50 per cohort, disjoint from the $N = 1{,}000$ evaluation traces): quality parameters are fitted via constrained nonlinear least-squares (achieving out-of-sample goodness-of-fit $R^2 = 0.964$ and RMSE $= 0.0048$, where empirical Flash Pass@1 scales monotonically from $93.0\%$ at $T_{\text{think}} = 0$ to $95.2\%$ at $T_{\text{think}} = 8$, $96.4\%$ at $T_{\text{think}} = 16$, and $98.8\%$ at $T_{\text{think}} = 128$, while Argon's surrogate intercept $Q_{\text{base}} = 1.02$ ensures worst-case compacted cyclic Cohort D prompts with $\delta_{\text{D}} + \delta_{\text{inst}} = 0.065$ evaluate to $1.02 - 0.065 = 0.955 \geq Q^* = 0.95$), $\alpha_{\text{prefill}} = 0.04$ ms/tok is fitted via linear regression of prefill TTFT against effective uncached tokens across both tiers ($R^2 = 0.941$), $\hat{L}_{\text{sandbox},k}$ is the sample mean sandbox verification latency on $N_{\text{cal}}$, and $\hat{L}_{\text{out}}(M, x, c)$ is set to the P95 single-turn core output token quantile on $N_{\text{cal}}$ (excluding the optional `schema_audit` diagnostic trailer—a structured post-payload audit block listing verified schema table/column bindings, nullability guards, and parameter role mappings consumed by downstream observability loggers and stripped prior to SQL/API execution; when `schema_audit` is enabled under strict pre-dispatch SLA budgeting, $\hat{L}_{\text{out},k}$ in Eqs. 22–23 is incremented by the calibrated trailer allowance $\Delta\hat{L}_{\text{audit},k} \in \{145, 30, 25, 8, 12\}$ tokens across Cohorts A–E without altering any cohort's tier selection):

- **Cohort A (SQL DB):** $\delta_{\text{A}} = 0.000$, $\hat{L}_{\text{out,A}} = 64$, $\hat{L}_{\text{sandbox,A}} = 714.2$ ms, $L_{\max,\text{A}} = 2{,}000$ ms ($L^{\text{model}}_{\max,\text{A}} = 1{,}285.8$ ms).
- **Cohort B (OpenAPI Agent):** $\delta_{\text{B}} = 0.015$, $\hat{L}_{\text{out,B}} = 96$, $\hat{L}_{\text{sandbox,B}} = 655.0$ ms, $L_{\max,\text{B}} = 2{,}000$ ms ($L^{\text{model}}_{\max,\text{B}} = 1{,}345.0$ ms).
- **Cohort C (Financial Compliance):** $\delta_{\text{C}} = 0.020$, $\hat{L}_{\text{out,C}} = 112$, $\hat{L}_{\text{sandbox,C}} = 641.0$ ms, $L_{\max,\text{C}} = 2{,}000$ ms ($L^{\text{model}}_{\max,\text{C}} = 1{,}359.0$ ms).
- **Cohort D (Codebase Refactoring):** $\delta_{\text{D}} = 0.035$, $\hat{L}_{\text{out,D}}(M, x, c = 0) = 256$ in uncompacted full-file mode (102 in format-matched unified-diff baseline mode), $\hat{L}_{\text{out,D}}(M, x, c = 1) = 94$ on acyclic instances ($\delta_{\text{inst}}(x) = 0$, where Stage 3 appends the 6-token directive `"Emit minimal unified-diff patches only."` under `### REQUIRED OUTPUT SPECIFICATION:`) and $\hat{L}_{\text{out,D}}(M, x, c = 1) = 256$ for all $M \in \mathcal{M}$ on cyclic instances ($\delta_{\text{inst}}(x) = 0.030$, which require full-module JTA/interface refactoring), $\hat{L}_{\text{sandbox,D}} = 482.0$ ms, $L_{\max,\text{D}} = 2{,}500$ ms ($L^{\text{model}}_{\max,\text{D}} = 2{,}018.0$ ms).
- **Cohort E (Multi-Turn Reasoning, per-turn):** $\delta_{\text{E}} = 0.010$, $\hat{L}_{\text{out,E}} = 80$, $\hat{L}_{\text{sandbox,E}} = 662.0$ ms, $L_{\max,\text{E}} = 2{,}000$ ms ($L^{\text{model}}_{\max,\text{E}} = 1{,}338.0$ ms per turn).

At runtime, a deterministic priority-ordered AST classifier (Cohort E $\succ$ D $\succ$ C $\succ$ B $\succ$ A to resolve hybrid multi-domain prompts unambiguously) evaluates five exact boolean predicates in $< 0.15$ ms: IsCohortE($x$) $\iff$ `r"(retry_count|HTTP\s+412|ETag\s+mismatch)"` in headers; IsCohortD($x$) $\iff$ `r"\b(package\s+[a-z0-9_.]+|import\s+javax\.transaction|func\s+\w+\(|def\s+\w+\(|from\s+[a-z0-9_.]+\s+import)\b"` (covering Java, Go, and Python repository refactoring); IsCohortC($x$) $\iff$ `r"\b(FinCEN|SOX|31\s+CFR|AML|suspicious_activity|beneficiary_ledger)\b"`; IsCohortB($x$) $\iff$ AST root contains JSON Schema/OpenAPI keys {`"openapi"`, `"paths"`, `"parameters"`, `"properties"`}; and IsCohortA($x$) $\iff$ `r"\b(SELECT|CREATE\s+TABLE|ALTER\s+TABLE)\b"`. Within the assigned cohort, Cohort A read-only queries are structurally fixed to $\delta_{\text{inst}}(x) = 0$ (as compacted read-only Cloud Spanner queries on $N_{\text{cal}}$ exhibited $< 0.4\%$ accuracy variation across join counts), whereas in Cohorts B–E the classifier assigns a shared cyclicity penalty $\delta_{\text{inst}}(x) = 0.030$ (constrained as a single shared scalar parameter during calibration because unrestricted per-cohort least-squares fits $\hat{\delta}_{\text{inst},k} \in \{0.029, 0.031, 0.033, 0.028\}$ across Cohorts B–E clustered tightly within $0.030 \pm 0.003$, inducing a uniform $\Delta\Omega(\text{Flash}) = +0.50$ shift [a $\sqrt{4{,}097} \approx 64\times$ budget jump] that prevents overfitting while cleanly handing off cyclic tasks to Argon) via four deterministic regex/AST checks: (1) Cohort B: JSON pointer graph contains a cycle ($\exists p \in$ `$ref` targeting an ancestor prefix) or recursive map `"additionalProperties":{"$ref":...}`; (2) Cohort C: compliance rule text or ledger DDL matches `r"(INTERLEAVE\s+IN\s+PARENT|round[_-]?trip|structuring\s+loop|cyclic[_-]?transfer|correspondent[_-]?chain)"`; (3) Cohort D: source payload contains $\geq 2$ cross-package/cross-module `import` or `from ... import` statements with bidirectional module qualifiers or matches `r"(UserTransaction\.\w+|setRollbackOnly|XAResource|__exit__|rollback)"`; and (4) Cohort E: state header matches `r"(retry_count\s*[:=]\s*[2-9]|HTTP\s+412.*HTTP\s+412)"` (defaulting conservatively to $\delta_{\text{default}} = 0.035$, $\hat{L}_{\text{out,default}} = 256$, $\hat{L}_{\text{sandbox,default}} = 600.0$ ms, $L_{\max,\text{default}} = 2{,}000$ ms for unseen workloads). For non-reasoning tiers ($\Delta Q_{\max}(M) = 0$), $T^*_{\text{think}}(M) = 0$ if $\text{Quality}(M, x, 0, c) \geq Q^*$, and tier $M$ is marked quality-infeasible otherwise. For reasoning-capable tiers ($\Delta Q_{\max}(M) > 0$), we define the normalized quality deficit:

$$\Omega(M) \triangleq \frac{Q^* - Q_{\text{base}}(M) + \delta_{\text{cohort}} + \delta_{\text{inst}}(x) + \delta_{\text{uncomp}}(1-c)}{\Delta Q_{\max}(M)} \tag{25}$$

For $\Omega(M) \leq 1$, the exact minimal integer thinking budget on $\{0, \ldots, T_{\max}\}$ is solved in closed form:

$$T^*_{\text{think}}(M) = \left\lceil \max\Big(0,\ (1 + T_{\max})^{\Omega(M)} - 1\Big) \right\rceil \tag{26}$$

Because $(1 + T_{\max})^{\Omega(M)}$ is exponential in $\Omega(M)$, calibration estimation errors $\pm\epsilon_\Omega$ induce a multiplicative budget shift $(1 + T_{\max})^{\pm\epsilon_\Omega}$, which is bounded above by the SLA latency ceiling $L^{\text{model}}_{\max,k}$ that automatically hands off high-deficit cyclic instances to Argon whenever $T^*_{\text{think}}(\text{Flash})$ becomes latency-prohibitive. If $\Omega(M) > 1$ or $\text{Latency}(M, L_{\text{in}}, L_{\text{prefix}}, \hat{L}_{\text{out}}(M, x, c), T^*_{\text{think}}(M), h^{(M)}) > L^{\text{model}}_{\max,k}$, tier $M$ is marked infeasible; when no tier satisfies both constraints simultaneously, the router applies an explicit two-stage fallback: (i) if the quality-feasible set $\mathcal{M}_{\text{qual}} = \{M \in \mathcal{M} : \text{Quality}(M, x, T^*_{\text{think}}(M), c) \geq Q^*\}$ is non-empty but all $M \in \mathcal{M}_{\text{qual}}$ exceed $L^{\text{model}}_{\max,k}$ under conservative P95 admission parameters—as occurs on cyclic Cohort D instances ($c = 1, \delta_{\text{inst}} = 0.030$), where Flash is quality-infeasible ($\Omega(\text{Flash}) = 0.065/0.060 > 1 \implies \mathcal{M}_{\text{qual}} = \{\text{Argon}\}$) and Argon's conservative P95 pre-dispatch latency ($850 + 0.04L_{\text{in,eff}} + 10^3(256)/78 \approx 4{,}132$ ms $+ 0.04L_{\text{in,eff}}$) exceeds $L^{\text{model}}_{\max,\text{D}} = 2{,}018.0$ ms—the router selects $M^* = \arg\min_{M \in \mathcal{M}_{\text{qual}}} \text{Latency}(M, \ldots) =$ Argon via fallback rule (i) while emitting a pre-dispatch P95 SLA-exceedance warning (whereas under empirical mean production throughput $\overline{\text{TTFT}}_0^{\text{Argon}} \approx 420$ ms and $\overline{\text{TPS}}^{\text{Argon}} \approx 145$ tok/s, realized Cohort D model latency averages 1,064.6 ms $\leq$ 2,018.0 ms); (ii) if $\mathcal{M}_{\text{qual}} = \emptyset$ (as occurs in the $c = 0$ uncompacted ablation on cyclic Cohort D where $\text{Quality}(\text{Argon}, x, 0, c = 0) = 1.02 - 0.035 - 0.030 - 0.015 = 0.940 < 0.95$), the router selects the maximum-quality tier $M^* = \arg\max_{M \in \mathcal{M}} \text{Quality}(M, x, T_{\max}, c)$ (Argon). Evaluating Eq. (26) across $|\mathcal{M}| = 2$ base tiers selects the optimal configuration in $< 0.1$ ms.

## IV. System Architecture and Implementation

The `PromptVM` architecture comprises two co-designed components (Fig. 1): (1) an in-process client-side Python/C++ SDK middleware compatible with the Google GenAI SDK (`google-genai`) on Google Cloud Vertex AI (with an optional FastAPI sidecar wrapper for polyglot microservices; in horizontally scaled multi-replica sidecar deployments or under high-concurrency TPU v6e HBM2e LRU eviction pressure, the $(M, \text{SHA-256}(\mathcal{P}_{\text{prefix}}))$ cache-state table is backed by a shared Redis/Valkey store and synchronized against RPC response headers—`cached_content_token_count` on Vertex AI and `x-tpu-kv-cache-hit` on TPU v6e—to eliminate cross-replica false negatives and LRU-eviction false positives), which intercepts outbound request payloads, executes Stages 1–3 compaction/anchoring, and solves Stage 5 Pareto routing; and (2) for self-hosted Cloud TPU v6e Trillium deployments, a custom server-side JAX/Pallas RPC handler and Block-Attention kernel that ingests pre-tokenized static prefix arrays with Stage 4 interior padding bitmasks ($\mathcal{I}_{\text{pad}}$) and decoupled logical RoPE/window counters.

Formally, the end-to-end compilation pipeline exhibits linear time and space complexity in prompt length: Stage 1 executes in $\mathcal{O}(|\mathcal{T}_{\text{AST}}| \log d_{\max} + |S|)$ time (via bottom-up Merkle-style sub-tree hashing) and $\mathcal{O}(|S|)$ space (where $d_{\max}$ is the maximum object key degree and $|\mathcal{S}_{\text{dense}}| \leq 8$); Stage 2 executes in $\mathcal{O}(|I|)$ time and $\mathcal{O}(|\mathcal{S}_{\text{seen}}| + |I|) = \mathcal{O}(|I|)$ space via the pre-compiled Aho-Corasick automaton and hash-set signature table; Stage 3–4 framing executes in $\mathcal{O}(|I| + |S|)$ time and $\mathcal{O}(B)$ auxiliary buffer space; and Stage 5 evaluates Eq. (26) in $\mathcal{O}(|\mathcal{M}|) = \mathcal{O}(1)$ time and $\mathcal{O}(1)$ space, yielding overall complexity $\mathcal{O}(|I| + |S| \log d_{\max})$ time and $\mathcal{O}(|I| + |S|)$ space. Benchmarking in-process client SDK compilation overhead across $N = 1{,}000$ traces confirms a median total latency of 1.80 ms decomposed across the median trace as Stage 1: 0.75 ms, Stage 2: 0.38 ms, Stage 3–4 framing: 0.42 ms, and Stage 5 router: 0.25 ms (with arithmetic mean total latency 1.92 ms and P95 3.20 ms; adding 0.60 ms loopback IPC when deployed as an external FastAPI sidecar), which is negligible relative to cloud inference latencies ($> 1{,}000$ ms).

## V. Empirical Evaluation

We conducted an empirical benchmark across **5 independent random seeds** ($s \in \{42, 43, 44, 45, 46\}$) at deterministic decoding temperature $\tau_{\text{temp}} = 0.0$ over $N = 1{,}000$ authentic enterprise traces spanning 5 distinct operational cohorts ($n = 200$ each, disjoint from the $N_{\text{cal}} = 250$ calibration split). Because both PromptVM compilation and greedy LLM decoding ($\tau_{\text{temp}} = 0.0$) are deterministic on any fixed trace, the 5 seeds control (i) stratified bootstrap sampling of $N = 1{,}000$ evaluation traces (200 per cohort) from a 5,000-trace enterprise production pool and (ii) randomized request arrival-order permutations within each cohort, which govern dynamic TTL context-cache state $h^{(M)} \in \{0, 1\}$. Tables I–III report exact realizations for the median seed ($s = 42$, where $3/200 = 1.50\%$ of Cohort D traces escalate to Argon, yielding $99.70\%$ pre-escalation Pass@1), while across all 5 seeds (1,000 Cohort D trace-seeds) the mean pre-escalation Pass@1 is $99.68\% \pm 0.08\%$ ($16/1{,}000 = 1.60\%$ Cohort D trace-seed escalations, i.e., $3.2/200$ per seed), inter-seed token SD is $\leq 4.2$ tokens, and latency SD is $\leq 48.5$ ms:

- **Cohort A (SQL DB):** High-performance Cloud Spanner read-only query synthesis.
- **Cohort B (OpenAPI Agent):** Multi-tool autonomous agent function dispatch with strict parameter contracts.
- **Cohort C (Financial Compliance):** SOX and AML cross-border treasury ledger audit extraction.
- **Cohort D (Codebase Refactoring):** Large enterprise Java JTA to Go service modernization.
- **Cohort E (Multi-Turn Reasoning):** Dynamic tool error recovery and self-correction loops.

### A. Evaluation Methodology and Verification Oracles

To ensure complete transparency across multi-turn agentic workflows, we report three distinct accuracy metrics under identical deterministic verification oracles: (1) *Turn-1 Single-Call Pass@1 (Pre-Retry)*, measuring strict first-generation

TABLE I

EMPIRICAL TELEMETRY AND PERFORMANCE MATRIX ACROSS $N = 1{,}000$ ENTERPRISE TRACES (5 SEEDS, MEDIAN SEED $s = 42$, $\tau_{\text{TEMP}} = 0.0$; SEPARATING MANAGED VERTEX AI ENDPOINTS FROM SELF-HOSTED CLOUD TPU V6E TRILLIUM GEMMA-2-9B MEASUREMENTS)

| Evaluation Dimension & Execution Regime | Monolithic Baseline (Gemini 4.0 Argon) | Naive Regex Trimming (on Argon) | Tokenomics Compiler (PromptVM) | Certified Impact |
|---|---|---|---|---|
| ***Regime A: Managed Google Cloud Vertex AI Endpoints*** *(Gemini 3.8 Flash / Flash-Thinking / 4.0 Argon; Stage 4 Padding Bypassed < 32k)* | | | | |
| Mean Raw Input Tokens | 466.7 tokens | 466.7 tokens | **466.7 tokens** | Standardized Baseline |
| Mean Compacted Input Tokens | 466.7 tokens (0.00% red) | 424.7 tokens (9.00% red) | **384.9 tokens (17.52% red)** | **17.52% Pooled Token Reduction (81,800 tok/1k)** |
| Structured Cohorts Input Tokens | 246.3 tokens (0.00% red) | 218.4 tokens (11.33% red) | **147.9 tokens (39.98% red)** | **39.98% Pooled Reduction (Cohorts A, B, C, E)** |
| Total Incurred Workflow Cost (1k req) | $1.7608 ($1.4568 diff-matched‡) | $1.5580 | **$1.0663 ($0.9812 core-only)** | **39.44% Net Savings (44.28% Core-Only)** |
| Maximum Cohort Cost Savings | Baseline (0.00%) | 18.20% (Cohort A: SQL) | **88.79% (Cohort A: SQL)** | **Near-Order-of-Magnitude Reduction** |
| Mean End-to-End Latency (Per-Turn / Cum Wf) | 2,450.0 ms/t (3,318.0 ms wf; 2,198.4 diff‡) | 2,180.0 ms/t | **1,489.4 ms/t (2,068.4 ms wf)** | **39.21% Per-Turn Red. (37.66% Cum Wf)** |
| P95 End-to-End Workflow Latency | 2,850.0 ms | 2,420.0 ms | **1,742.8 ms** | **38.85% Baseline Latency Reduction** |
| Turn-1 Single-Call Pass@1 (Pre-Retry)† | 85.00% | 78.40% | **99.70% (99.68% 5-seed mean)** | **+14.70% Turn-1 Accuracy Gain ($Q^* \geq 0.95$)** |
| Post-Retry Same-Tier Pass@k (Pre-Escalation)† | 98.50% | 89.20% | **99.70%** | **+1.20% Over Multi-Turn Monolithic Argon** |
| Post-Verification Workflow Completion Rate† | 98.50% | 89.20% | **100.00%** | **Zero Unresolved Task Failures** |
| ***Regime B: Self-Hosted Cloud TPU v6e Trillium Cluster*** *(Open-Weights Gemma-2-9B + Custom Server-Side JAX/Pallas Kernel; Stage 4 Active on Sub-32k Prompts)* | | | | |
| Attention Softmax Entropy ($\bar{N} = 466.7$) | 6.12 nats (diffuse) | 6.05 nats (1.14% red) | **4.18 nats (bounded)** | **31.70% Entropy Reduction ($\mathcal{H}_{\text{empirical}} \leq \mathcal{H}_{\max}$)** |
| Boundary-Page Sealing & Multi-Page Util. | Unsealed ($\Delta_{\text{frag}} > 0$, 91.40% util.) | Unsealed (91.40% util.) | **100.00% Sealed** ($k \cdot B, B = 16$) | **Zero CoW Boundary Duplication (-14.23% 8-Way TTFT)** |
| Unified Gemma-2-9B Task Accuracy (Turn-1 / Wf) | 91.20% / 95.40% | 83.50% / 87.10% | **96.40% / 98.90%** | **+5.20% Single-Model Pass@1 (-28.4% HBM-VMEM Traffic)** |

†Turn-1 Single-Call Pass@1 measures strict first-generation accuracy before same-tier retries ($\bar{R}_{\text{A}} = 1.35$ [65.0% Turn-1], $\bar{R}_{\text{D}} = 1.40$ [60.0% Turn-1] on Baseline); Post-Retry Same-Tier Pass@k measures success after same-tier retries; Post-Verification Wf includes cross-tier cascade escalation (16/1,000 Cohort D trace-seeds = 1.60% 5-seed mean [3.2/200; 3/200 = 1.50% on median seed $s = 42$] escalate to Argon).

‡Diff-matched baseline appends the identical 6-token unified-diff directive to Monolithic Argon on acyclic Cohort D traces, reducing Cohort D baseline cost from $5.9400 → $4.4200/1k ($1.7608 → $1.4568/1k pooled).

accuracy before any same-tier retry or cross-tier escalation; (2) *Post-Retry Same-Tier Pass@k (Pre-Escalation)*, measuring task accuracy after same-tier verification retries ($\bar{R}_{\text{A}} = 1.35$ in Cohort A and $\bar{R}_{\text{D}} = 1.40$ in Cohort D on Monolithic Baseline) prior to any cross-tier cascade fallback; and (3) *Post-Verification Workflow Completion Rate (Wf)*, measuring final workflow success including oracle-triggered cross-tier escalation to Argon. Across all cohorts, SQL queries are validated against live Cloud Spanner AST schemas and execution snapshots [13]; OpenAPI tool calls are verified against JSON Schema contracts [8] and parameter role/invariant assertions; compliance extractions are audited against FinCEN rule engines; code refactorings are verified by compiling generated Go source files (`go build`) and executing automated unit test suites verifying transactional rollback parity against Java JTA; and multi-turn reasoning is audited via a deterministic state machine asserting JSON patch convergence, ETag conflict resolution, and zero diagnostic exceptions across $\bar{R}_{\text{E}} = 3.0$ sequential turns. End-to-end workflow latency in Tables I–III sums client-perceived model RPC latency and empirical sandbox/oracle execution ($\hat{L}_{\text{sandbox},k} \in [482.0, 714.2]$ ms). Target quality is set to $Q^* = 0.95$ with default end-to-end SLA $L_{\max,k} = 2{,}000$ ms ($L_{\max,\text{D}} = 2{,}500$ ms for Cohort D).

### *B. Summary Telemetry, Live Vertex AI Golden Verification, and TPU v6e Hardware Validation*

Table I separates metrics collected on managed Vertex AI endpoints (Regime A) from white-box hardware telemetry on our self-hosted Cloud TPU v6e Trillium Gemma-2-9B cluster (Regime B). All $N = 1{,}000$ evaluation traces in Regime A were executed live against Google Cloud Vertex AI endpoints (`gemini-2.5-flash` with dynamic `thinking_budget` and `gemini-2.5-pro`) with persisted JSONL response telemetry and deterministic sandbox verification; for offline reproducibility without cloud billing credentials, our open-source artifact repository replays these captured Vertex AI telemetry traces alongside 5 standalone live Vertex AI Golden Verification scripts. Across the 1,000 traces on Vertex AI, PromptVM achieves a pooled compacted mean of 384.9 tokens (17.52% global token reduction, saving 81,800 tokens per 1k requests; 39.98% pooled reduction on structured Cohorts A, B, C, and E), a 39.44% pooled invoice cost reduction ($1.0663 vs. $1.7608 per 1k requests; 40.22% unweighted mean per-cohort cost reduction across the 5 cohorts; 43.47% pooled cost reduction to $0.9953/1k when disabling `schema_audit` across Cohorts A–C alone, or 44.28% pooled reduction to $0.9812/1k when disabling optional audit and diagnostic trailers across all five cohorts A–E; and 26.81% vs. the $1.4568 diff-matched baseline), a 99.70% Turn-1 Single-Call Pass@1 (vs. 85.00% Turn-1 Pass@1 and 98.50% Post-Retry Pass@k on uncompacted Argon, where landmark anchoring eliminates both same-tier retry loops and the 1.50% attention-dispersion constraint omissions), and 100.00% post-verification workflow completion. Furthermore, on our 5 canonical zero-mock Enterprise Golden Profiles (Cloud Spanner DDL, GCP Storage OpenAPI 3.1, FinCEN 31 CFR 1010.311 AML, Java JTA to Go Modernization, and ETag HTTP 412 Recovery) dispatched live to Google Cloud Vertex AI (`project="gcp-enterprise-eval-prod"`, `model="gemini-2.5-flash"`), PromptVM reduced physical Vertex AI `prompt_token_count` by **27.74%** (2,098 → 1,516 tokens across the 5 golden tasks, and by **59.88%** [815 → 327 tokens] on the GCP Storage OpenAPI 3.1 specification) with 100% JSON schema validity. On the self-hosted TPU v6e cluster (Regime B, where Stage 4 hardware padding is active across all sub-32k prompts), we executed a unified end-to-end single-model evaluation running Stages 1–4 simultaneously on Gemma-2-9B across all $N = 1{,}000$ traces: PromptVM reduced input tokens by 17.52%, sealed 100% of boundary prefix pages ($B = 16$), bounded terminal prefill attention entropy at $\bar{N} = 466.7$ from $\mathcal{H}_{\text{diffuse}} = 6.12$ nats to $\mathcal{H}_{\text{empirical}} = 4.18$ nats ($\leq \mathcal{H}_{\max} \approx 4.28$ nats), reduced HBM2e-to-VMEM KV memory transfer latency by 28.4%, and lifted Gemma-2-9B Turn-1 Pass@1 from 91.20% → 96.40% (95.40% → 98.90% post-verification workflow completion). In our 8-way concurrent subagent fan-out benchmark on a single TPU v6e chip ($B = 16$) sharing a 134-token static system/schema prefix ($\Delta_{\text{frag}} = 10$, padded to $144 = 9 \times 16$ tokens) followed by 64 dynamic suffix tokens per branch, sealing the 9th boundary block as read-only eliminated 8-way CoW trailing-

block duplication and reduced batch prefill TTFT by $14.23\%$ ($41.6$ ms $\rightarrow$ $35.7$ ms).

### *C. Cohort Disaggregation and Analytical-to-Empirical Reconciliation*

Table II reports disaggregated telemetry alongside the exact per-cohort parameters that reconcile the single-pass analytical router (Eqs. 22–23) with multi-turn workflow billing: SLA budget $\hat{L}_{\text{out},k}$ vs. empirical mean realized output tokens per turn ($\bar{L}_{\text{out}}^{\text{real}}$ for both Baseline and PromptVM), mean same-tier verification turns ($\bar{R}_k$, baseline $\rightarrow$ ours, separated from cross-tier escalation $p_{\text{esc}}$), turn-averaged context cache hit fraction ($\bar{h}$, reported as Base / Ours), Flash/Argon routing split, and separate model RPC vs. sandbox verification latency components (where `Model` latency equals server execution $\overline{\text{TTFT}}_0^{(M)} + \alpha_{\text{prefill}} L_{\text{in,eff}} + 10^3(\bar{L}_{\text{out}}^{\text{real}} + T_{\text{think}}^*)/\overline{\text{TPS}}^{(M)}$ plus regional Vertex AI gRPC/TLS framing and structured-output grammar-binding overhead $\Delta t_{\text{rpc}} \in [13.5, 43.2]$ ms). Specifically:

- **Cohort A (SQL DB):** Compaction reduces input tokens by 42.84% ($273.1 \rightarrow 156.1$). Baseline Argon averages $\bar{R}_{\text{A}} = 1.35$ execution-verification turns ($65.00\%$ Turn-1 Pass@1, $\bar{L}_{\text{out}}^{\text{real}} = 64.0$ tok/turn, $\bar{h} = 0.01$ with cached prefix $L_{\text{prefix}} = 180.0$ tok), where the plug-in mean formula evaluates to $1.35 \times [((0.99 \times 1.25 + 0.01 \times 0.3125) \times 180.0 + 1.25 \times 93.1 + 10.00 \times 64.0) \times 10^{-3}] = 1.35 \times \$0.97969 \approx \$1.3226/1\text{k}$ (and $\$1.3228/1\text{k}$ from the exact unrounded trace-level sum due to positive covariance between prompt length and retry count $\bar{R}_{\text{A}}$; routing uncompacted Cohort A prompts directly to Flash at $\bar{R}_{\text{A}} = 1.30$ costs $\$0.2185/1\text{k}$, isolating $32.1\%$ additional cost savings from Stages 1–3 compaction beyond tier selection alone). PromptVM routes $100\%$ of queries to Flash ($T_{\text{think}}^* = 0$, $\bar{R}_{\text{A}} = 1.00$ turn with zero retries, $\bar{h} = 0.00$, and $\bar{L}_{\text{out}}^{\text{real}} = 208.2$ tok when the optional multi-tier `schema_audit` diagnostic trailer—a structured post-payload comment block verifying schema table/column bindings, nullability constraints, and join-key lineage for enterprise audit logging that is stripped prior to Cloud Spanner execution—is enabled), reducing cost to $\$0.1483/1\text{k}$ (88.79% savings) and end-to-end latency to $1{,}545.4$ ms ($831.2$ ms model RPC + $714.2$ ms Spanner sandbox; when the `schema_audit` trailer is disabled to match Baseline's SQL-only output $\bar{L}_{\text{out}}^{\text{real}} = 61.8$ tok, Cohort A cost drops to $\$0.0605/1\text{k}$ [95.43% savings] and model RPC latency drops by $457.5$ ms to $373.7$ ms, yielding $1{,}087.9$ ms total latency).
- **Cohorts B & C (OpenAPI & Financial Compliance):** Because baseline realized output payloads are compact JSON function calls ($\bar{L}_{\text{out}}^{\text{real}} = 11.3$ tok in B) and structured compliance verdicts ($\bar{L}_{\text{out}}^{\text{real}} = 9.5$ tok in C) at $\bar{R}_k = 1.00$, single-pass Argon baseline costs are $274.2 \times \$0.00125 + 11.3 \times \$0.010 = \$0.4558/1\text{k}$ at $\bar{h} = 0.00$ ($\$0.4555/1\text{k}$ exact trace sum including $\bar{h} = 0.002$ prefix cache hits on $L_{\text{prefix}} = 160.0$ tok, with $38.99\%$ token reduction from $274.2 \rightarrow 167.3$ tok) in B, and $236.0 \times \$0.00125 + 9.5 \times \$0.010 = \$0.3900/1\text{k}$ in C. Under PromptVM, acyclic traces in Cohort B ($72.0\%$, $\Omega(\text{Flash}) = 0.015/0.060 = 1/4$) allocate $T_{\text{think}}^* = \lceil 4{,}097^{1/4} - 1 \rceil = \mathbf{8}$ tokens ($\bar{L}_{\text{in}}^{\text{Flash}} = 84.2$ tok, $\bar{L}_{\text{out}}^{\text{Flash}} = 32.2$ tok with trailer [$11.3$ tok without]), and acyclic traces in Cohort C ($72.5\%$, $\Omega(\text{Flash}) = 0.020/0.060 = 1/3$) allocate $T_{\text{think}}^* = \lceil 4{,}097^{1/3} - 1 \rceil = \lceil 15.0005 \rceil = \mathbf{16}$ tokens ($\bar{L}_{\text{in}}^{\text{Flash}} = 110.6$ tok, $\bar{L}_{\text{out}}^{\text{Flash}} = 28.6$ tok with trailer [$9.5$ tok without]). Conversely, multi-file cyclic schemas ($\delta_{\text{inst}} = 0.030$, $28.0\%$ in B and $27.5\%$ in C) require $\Omega(\text{Flash}) = 0.75 \implies T_{\text{think}}^* = 512$ in B and $\Omega(\text{Flash}) = 5/6 = 0.83\overline{3} \implies T_{\text{think}}^* = 1{,}024$ in C, whose conservative P95 model latencies ($3{,}618$ ms and $6{,}470$ ms) exceed $L_{\max,k}^{\text{model}}$ ($1{,}345.0$ ms and $1{,}359.0$ ms) and therefore trigger pre-dispatch routing to Argon ($\bar{L}_{\text{in}}^{\text{Argon}} = 381.0$ tok, $\bar{L}_{\text{out}}^{\text{Argon}} = 61.5$ tok in B [$11.3$ tok without trailer]; $\bar{L}_{\text{in}}^{\text{Argon}} = 341.2$ tok, $\bar{L}_{\text{out}}^{\text{Argon}} = 49.2$ tok in C [$9.5$ tok without trailer], satisfying the exact cohort-wide input means $0.72(84.2) + 0.28(381.0) = 167.3$ tok and $0.725(110.6) + 0.275(341.2) = 174.0$ tok and output means $40.4$ tok and $34.3$ tok), yielding exact workflow costs of $0.72(0.0842 \times 0.15 + 0.0322 \times 0.60 + 0.008 \times 0.60) + 0.28(0.3810 \times 1.25 + 0.0615 \times 10.00) = \$0.3320/1\text{k}$ (27.11% savings; $\$0.1824/1\text{k}$ [59.96% savings] with `schema_audit` disabled) in B and $0.725(0.1106 \times 0.15 + 0.0286 \times 0.60 + 0.016 \times 0.60) + 0.275(0.3412 \times 1.25 + 0.0492 \times 10.00) = \$0.2840/1\text{k}$ (27.18% savings; $\$0.1665/1\text{k}$ [57.31% savings] with `schema_audit` disabled) in C.
- **Cohort D (Codebase Refactoring):** Prompts contain dense Java source payloads ($1{,}348.0 \rightarrow 1{,}333.0$ tok, $1.11\%$ compaction; shared repository context prefix $L_{\text{prefix}} = 640.0$ tok). Baseline Argon averages $\bar{R}_{\text{D}} = 1.40$ compile turns ($60.00\%$ Turn-1 Pass@1, $92.50\%$ Post-Retry Pass@k, turn-1 cache hit fraction $\bar{h}_{\text{Turn1}} = 0.18$ rising to turn-weighted $\bar{h}_{\text{Base}} = 0.270$ across retry turns as same-trace retries hit the warm $640.0$-tok repository prefix, and $\bar{L}_{\text{out}}^{\text{real}} = 266.6$ tok/turn effective billed output [$256.0$ tok Turn-1 output plus $10.6$ tok amortized retry diagnostic header]), costing $1.40 \times [((0.73 \times 1.25 + 0.27 \times 0.3125) \times 640.0 + 1.25 \times 708.0 + 10.00 \times 266.6) \times 10^{-3}] = 1.40 \times \$4.1890 + \$0.0754$ (trace covariance) $= \$5.9400/1\text{k}$ (equivalently $1.40 \times \$4.2430 = \$5.9400/1\text{k}$ when evaluated at the Turn-1 cache rate $\bar{h}_{\text{Turn1}} = 0.18$ with $266.6$ tok/turn, or $\$4.1890$/turn at $261.2$ tok/turn; under the diff-matched baseline where Argon also receives the 6-token unified-diff directive on the $34.0\%$ acyclic files—with acyclic sub-cohort parameters $\bar{L}_{\text{in}} = 1{,}126.0$ tok, $L_{\text{prefix}} = 640.0$ tok, $\bar{h} = 0.24$, $\bar{L}_{\text{out}}^{\text{real}} = 102.0$ tok/turn, and $\bar{R}_{\text{D,acyc}} = 1.25$ costing $\$1.4712/1\text{k}$ and blending with the $66.0\%$ cyclic sub-cohort at $\$5.9400/1\text{k}$ [$\bar{R}_{\text{D,cyc}} = 1.40$] to $0.34(\$1.4712) + 0.66(\$5.9400) = \$4.4206/1\text{k} \approx \$4.4200/1\text{k}$ at cohort-wide $\bar{R}_{\text{D}} = 1.35$ and pooled $\bar{R}_k = 1.52$—or $\$3.1312/1\text{k}$ when applied across all

100% of Cohort D traces at $\bar{L}_{\text{out}}^{\text{real}} = 118.4$ tok/turn and $\bar{R}_{\text{D}} = 1.30$). Under PromptVM, acyclic single-package files (34.0% of traces, $\delta_{\text{inst}} = 0$, $\bar{L}_{\text{in}}^{\text{Flash}} = 1{,}120.0$ tok, $L_{\text{prefix}} = 640.0$ tok, $\bar{h}^{\text{Flash}} = 0.35$) evaluate Eq. (26) with $\Omega(\text{Flash}) = 0.035/0.060 = 7/12 = 0.58\overline{3}$, yielding $\lceil (1 + 4{,}096)^{7/12} - 1 \rceil = \lceil 127.018 \rceil = \mathbf{128}$ tokens and emitting concise unified-diff patches ($\bar{L}_{\text{out}}^{\text{Flash}} = 94.0$ tok, costing \$0.2738/1k per Flash trace), while multi-file cyclic ASTs (66.0% of traces, $\delta_{\text{inst}} = 0.030 \implies \Omega(\text{Flash}) = 0.065/0.060 > 1$, $\bar{L}_{\text{in}}^{\text{Argon}} = 1{,}442.7$ tok, $L_{\text{prefix}} = 640.0$ tok, $\bar{h}^{\text{Argon}} = 0.5015$ [blended $\bar{h} = 0.45$], $\bar{L}_{\text{out}}^{\text{Argon}} = 434.2$ tok, cohort-wide blended $\bar{L}_{\text{out}}^{\text{real}} = 318.5$ tok [311.2 tok core code excluding a 7.3-tok verification trailer], costing \$5.8453/1k per Argon trace) route pre-dispatch to Argon. Same-tier turns average $\bar{R}_{\text{D}} = 1.00$ with Turn-1 Pass@1 of 98.50% (197/200); adding the $p_{\text{esc}} = 3/200 = 0.015$ post-compile `go build` escalations to cold Argon ($h^{(\text{Argon})} = 0, \bar{L}_{\text{in}}^{\text{esc}} = 1{,}120.0$ tok, $\bar{L}_{\text{out}}^{\text{esc}} = 548.0$ tok, adding $0.015 \times \$6.8800 = \$0.1032/1\text{k}$) yields $0.34(\$0.2738) + 0.66(\$5.8453) + \$0.1032 = \$4.0542/1\text{k}$ (or \$4.0473/1k core-only without the 7.3-tok diagnostic trailer; 31.75% savings vs. verbose baseline; 8.28% savings on Cohort D alone and 26.81% pooled savings vs. diff-matched Argon) and 100.00% workflow completion at 1,546.6 ms total latency (1,064.6 ms model RPC + 482.0 ms `go build`).

- **Cohort E (Multi-Turn Reasoning):** Across $\bar{R}_{\text{E}} = 3.00$ sequential state-machine turns with session prefix caching ($\bar{h} = 0.42$ across turns; Table II reports both per-turn `[/t]` and 3-turn cumulative workflow `[cum]` token totals alongside 3-turn workflow cost and per-turn mean latency 1,447.5 ms/t, which corresponds to $3 \times 1{,}447.5 = 4{,}342.5$ ms cumulative 3-turn workflow wall-clock latency), 53.47% input compaction (67.3 → 31.3 tok/turn, i.e., 202.0 → 94.0 tok cumulative; baseline cumulative prefix $L_{\text{prefix,cum}} = 144.0$ tok, core patch output $\bar{L}_{\text{out,cum}}^{\text{Base}} = 44.3$ tok [14.8 tok/turn] plus 5.69 tok [1.90 tok/turn] state-header echo [50.0 tok cumulative / 16.7 tok/turn total billed], costing $((0.58 \times 1.25 + 0.42 \times 0.3125) \times 144.0 + 1.25 \times 58.0 + 10.00 \times 50.0) \times 10^{-3} = \$0.6957/1\text{k}$) combined with Flash routing on acyclic traces (74.0% Flash at $T_{\text{think}}^{*} = \lceil 4{,}097^{1/6} - 1 \rceil = \mathbf{4}$ tok/turn [12 tok cum.], cumulative $\bar{L}_{\text{in,cum}}^{\text{Flash}} = 72.0$ tok, $L_{\text{prefix,cum}} = 54.0$ tok, $\bar{h}^{\text{Flash}} = 0.42$, $\bar{L}_{\text{out,cum}}^{\text{Flash}} = 54.0$ tok, costing \$0.0478/1k) and Argon routing on cyclic lock traces (26.0% Argon, where $\delta_{\text{inst}} = 0.030 \implies \Omega(\text{Flash}) = 2/3 \implies T_{\text{think}}^{*} = 256$ yields P95 model latency $320.2 + 10^3(80 + 256)/185 = 2{,}136.4$ ms [and 1,704.0 ms pre-output thinking completion], both exceeding $L_{\max,\text{E}}^{\text{model}} = 1{,}338.0$ ms; cumulative $\bar{L}_{\text{in,cum}}^{\text{Argon}} = 156.6$ tok, $L_{\text{prefix,cum}} = 114.0$ tok, $\bar{h}^{\text{Argon}} = 0.42$, $\bar{L}_{\text{out,cum}}^{\text{Argon}} = 166.9$ tok [124.0 tok core patch output excluding 42.9 tok of state-audit/echo trailers across 3 turns, of which 24.5 tok is the optional Argon state-audit trailer], satisfying the cohort-wide 3-turn cumulative input mean $0.74(72.0) + 0.26(156.6) = 94.0$ tok, total output $0.74(54.0) + 0.26(166.9) = 83.4$ tok [27.8 tok/turn], and core patch output $0.74(54.0) + 0.26(124.0) = 72.2$ tok [24.1 tok/turn], costing \$1.8370/1k) reduces 3-turn workflow cost to $0.74(\$0.0478) + 0.26(\$1.8370) = \$0.5130/1\text{k}$ (26.26% savings; or \$0.4493/1k [35.42% savings] when stripping the 24.5-tok Argon state-audit trailer [$0.26 \times 24.5$ tok $\times$ \$0.010 = \$0.0637/1k]).

Across all 5 equal-sized cohorts, Tables I–II aggregate token and cost columns on a **per-workflow (`wf / cum`)** basis while reporting both **per-invocation (`/t`)** and **cumulative per-workflow (`wf`)** latency (1,489.4 ms/t and 2,068.4 ms wf on PromptVM vs. 2,450.0 ms/t and 3,318.0 ms wf on Monolithic Baseline): the pooled baseline cost is $\frac{1}{5}(1.3228 + 0.4555 + 0.3900 + 5.9400 + 0.6957) = \$1.7608$ and the pooled PromptVM cost is $\frac{1}{5}(0.1483 + 0.3320 + 0.2840 + 4.0542 + 0.5130) = \$1.0663$ (39.44% pooled invoice savings; 40.22% unweighted mean of the five per-cohort percentage reductions; \$0.9953/1k [43.47% pooled savings] when disabling `schema_audit` across Cohorts A–C alone; and $\frac{1}{5}(0.0605 + 0.1824 + 0.1665 + 4.0473 + 0.4493) = \$0.9812/1\text{k}$ [44.28% pooled savings] with pooled per-workflow core output $\frac{1}{5}(61.8 + 11.3 + 9.5 + 311.2 + 72.2) = 93.2$ tok wf [83.6 tok/t] when optional audit and diagnostic trailers are disabled across all five cohorts A–E).

### *D. Component Ablation, SOTA Comparison, and Public Benchmarks*

Table III isolates the contribution of each PromptVM stage against state-of-the-art compression and routing baselines under identical multi-turn verification rules (reporting input tokens, realized workflow output tokens and mean turns $\bar{L}_{\text{out}}^{\text{real}}/\bar{R}_k$, workflow cost, per-invocation latency, and strict Turn-1 / Wf accuracy; all single-tier Argon rows have the multi-tier `schema_audit` trailer disabled so output schemas match Monolithic Baseline). First, decomposing Stage 1 on fixed Gemini 4.0 Argon isolates its two sub-mechanisms: *Stage 1a: Key Sort + Union Fold + 4-Part Guard* compacts 466.7 → 411.2 tokens (−55.5 tok) and *Stage 1b: `$defs`/`$ref` Sub-Tree Hoisting* saves an additional −2.7 tokens pooled across all cohorts (−13.5 tok across Cohort B's nested OpenAPI schemas), yielding *Stage 1 Complete* at 408.5 tokens (−58.2 tok, output-normalized cost \$1.6881/1k [realized \$1.5642/1k], 85.00% Turn-1 / 98.50% Wf; note that stripping verbose input schema descriptions and inline example blocks in Stage 1a/1b also suppresses echoed SQL/JSON field comments in Argon's responses, reducing realized output tokens by 8.1–8.9 tok [74.3 → 66.2 and 65.4 tok] and trimming $\bar{R}_k$ from 1.55 → 1.51). *Stage 2 Only (Span-Strip + Invariant Pruning)* compacts 466.7 → 440.6 tokens (−26.1 tok, output-normalized \$1.7282/1k [realized \$1.5180/1k], 82.10% Turn-1 / 95.80% Wf without Stage 3 anchors), and combining Stages 1+2 while omitting only Stage 3's 7-token input landmark sink header (*Compaction w/o Landmark Anchors*: 382.4 in tok

TABLE II
DISAGGREGATED PERFORMANCE AND ANALYTICAL RECONCILIATION PARAMETERS BY ENTERPRISE COHORT ($n = 200$ EACH, 5 SEEDS, MEDIAN SEED $s = 42$)

| Cohort | Raw → Comp. Tok. | Tok. Red. | SLA $\bar{L}_{\text{out},k}$ / Base Real (Ours Real) | Turns $\bar{R}_k$ (Base→Ours) | Cache $\bar{h}$ (B/O) | Flash / Argon Split | Cost: Base → Ours (/1k) | Cost Sav. | Latency: Model + Sandbox = Total | Pass@1 (1st / Wf)† |
|---|---|---|---|---|---|---|---|---|---|---|
| Cohort A: SQL DB | 273.1 → 156.1 | 42.84% | 64 / 64.0 (208.2‡ [61.8 SQL]) | 1.35 → 1.00 | 0.01 / 0.00 | 100.0% / 0.0% ($T^*_{\text{think}} = 0$) | $1.3228 → $0.1483 ($0.0605‡) | 88.79% | 831.2 + 714.2 = 1,545.4 ms | 100.00% / 100.00% |
| Cohort B: OpenAPI Agent | 274.2 → 167.3 | 38.99% | 96 / 11.3 (40.4‡ [11.3 core]) | 1.00 → 1.00 | 0.00 / 0.00 | 72.0% / 28.0% ($T^*_{\text{think}} = 8$) | $0.4555 → $0.3320 ($0.1824‡) | 27.11% | 798.4 + 655.0 = 1,453.4 ms | 100.00% / 100.00% |
| Cohort C: Financial Compliance | 236.0 → 174.0 | 26.27% | 112 / 9.5 (34.3‡ [9.5 core]) | 1.00 → 1.00 | 0.00 / 0.00 | 72.5% / 27.5% ($T^*_{\text{think}} = 16$) | $0.3900 → $0.2840 ($0.1665‡) | 27.18% | 812.9 + 641.0 = 1,453.9 ms | 100.00% / 100.00% |
| Cohort D: Codebase Refactoring | 1,348.0 → 1,333.0 | 1.11% | 256 / 266.6 (318.5‡ [311.2 core]) | 1.40 → 1.00 (+0.015 esc) | 0.27 / 0.45 | 34.0% / 66.0% ($T^*_{\text{think}} = 128$) | $5.9400 ($4.4200 diff) → $4.0542 ($4.0473‡) | 31.75% | 1,064.6 + 482.0 = 1,546.6 ms | 98.50% / 100.00% |
| Cohort E: Multi-Turn Reasoning§ | 67.3→31.3/t (202→94 cum) | 53.47% | 80/t / 14.8/t (83.4‡ [72.2 core] cum) | 3.00 → 3.00 | 0.42 / 0.42 | 74.0% / 26.0% ($T^*_{\text{think}} = 4$/t) | $0.6957 → $0.5130 ($0.4493‡ cum) | 26.26% | 785.5+662.0=1,447.5/t (4,342.5 wf) | 100.00% / 100.00% |
| **Pooled Mean** ($N = 1{,}000$)§ | **466.7 → 384.9 (439.8→372.3/t)** | **17.52%** | **121.6 / 73.2 wf (134.7 wf [93.2 core]; 123.6/t)** | **1.55 → 1.40** | **0.14 / 0.17** | **70.5% / 29.5%** | **$1.7608 → $1.0663 ($0.9812‡)** | **39.44%** | **858.5+630.9=1,489.4/t (2,068.4 wf)** | **99.70% / 100.00%** |

†PromptVM Turn-1 Pass@1 / Post-Verification Wf shown (Baseline Turn-1 Pass@1 is 65.00% in A, 100.00% in B/C/E, 60.00% in D [92.50% Post-Retry], and 85.00% Pooled [98.50% Post-Retry]).
‡Disabling optional audit/diagnostic trailers across A–E yields core outputs (61.8, 11.3, 9.5, 311.2, 72.2 tok wf; 93.2 tok wf [83.6/t] pooled) and costs ($0.0605, $0.1824, $0.1665, $4.0473, $0.4493; $0.9812/1k pooled).
§Cohort E reports per-turn (/t) and 3-turn workflow (cum/wf) totals; Pooled Base Real 73.2 tok wf (67.3/t) is core output (74.3 tok wf [67.7/t] in Table III adds Cohort E's 5.69-tok cum [1.90/t] echo, +1.14 wf pooled; Ours 123.6/t [122.1/t w/o D trailer]).

and 42.1 out tok, reflecting $-58.2 - 26.1 = -84.3$ in tok before Stage 3 replaces 4.5 tok of informal labels with the 7.0-token header to reach 384.9 in tok, while retaining Stage 3's acyclic Cohort D unified-diff output format directive; omitting both the input landmark header and the unified-diff output directive yields 61.2 out tok at the identical 82.10% Turn-1 / 95.80% Wf accuracy) proves that Stage 3's $K = 7$ landmark sink header is the specific mechanism that lifts Turn-1 Pass@1 from 82.10% to 100.00% on Argon (*Compaction Only*). Second, single-tier compaction configurations on Argon (whose empirical mean production throughput is $\overline{\text{TPS}}^{\text{Argon}} = 145$ tok/s, i.e., 6.90 ms/output token) reduce end-to-end workflow latency by 470.0 ms (2,450.0 → 1,980.0 ms) primarily through two compounding effects beyond direct prefill savings ($\alpha_{\text{prefill}}\Delta L_{\text{in}} \approx 3.3$ ms): (i) Stage 3 landmark anchoring eliminates Turn-1 constraint omissions, cutting mean workflow turns $\bar{R}_k$ from 1.55 → 1.40 ($\Delta\bar{R}_k = -0.15$ avoided retry turns, saving $0.15 \times 1{,}680$ ms $= 252.0$ ms of retry prefill, decode, and RPC turnaround), and (ii) workflow output token reduction of −31.6 tok (74.3 → 42.7 tok, saving $31.6 \times 6.90$ ms $= 218.0$ ms of Argon decode time, which decomposes into −22.4 tok pooled from Stage 3's acyclic Cohort D unified-diff directive [matching the diff-matched baseline 74.3 → 51.9 tok] plus −9.2 tok from Stage 2 conversational preamble suppression [51.9 → 42.7 tok]). In Table III, the five Stage 1/2/3 isolation rows marked with § report output-normalized cost (fixing output tokens and turns to Baseline's 74.3 tok and $\bar{R}_k = 1.55$ to isolate pure input-token billing savings of $-\$0.10225/1\text{k}$ from 466.7 → 384.9 tokens, while listing actual realized multi-turn invoice costs alongside in parentheses, e.g., $\$1.2946/1\text{k}$ for *Compaction Only*), whereas in Table I, *Naive Regex Trimming* ($1.5580) saves $0.0525 on input tokens (−42.0 tok) plus $0.1503 from premature output truncation (−15.03 output tok) caused by broken schema clauses (crashing Turn-1 Pass@1 to 78.40% and Wf to 89.20%).

All statistical prompt-compression baselines in Table III—LLMLingua [17], [18], LLMLingua-2 [19], and Selective Context [20]—were evaluated on Gemini 4.0 Argon both at their standard $0.73\times$–$0.80\times$ retention ratios (342.0–373.3 tokens) and at an exact budget-matched 384.9 tokens (*LLMLingua-2 Matched Budget*): token-level perplexity pruning drops schema delimiters and column keys, degrading Wf accuracy to 74.20%–86.10%, and restricting LLMLingua strictly to unstructured instructions $I$ in *Schema-Safe LLMLingua + Stage 1 AST* still drops negative modal qualifiers `never`/`not` (81.20% Turn-1 / 94.60% Wf). Similarly, deterministic grammar-only schema minification and description stripping without our 4-part semantic guard (*Unguarded Schema Description Stripping*: 392.1 tokens, $1.5910/1k, 79.50% Turn-1 / 91.40% Wf) loses sibling parameter role disambiguation on Cohort B, whereas PromptVM's guarded Stage 1 preserves 100% schema contract fidelity. On the TPU v6e hardware plane (Regime B), comparing Stage 4 against modular position-independent KV-cache block reuse [23] confirms that while decoupled block caching eliminates cross-document recomputation, unaligned trailing prefix blocks still incur CoW duplication under 8-way subagent fan-out (40.8 ms TTFT vs. PromptVM's 35.7 ms with zero-initialized Pallas VMEM masking). To ensure a strictly fair comparison against standalone routers (FrugalGPT [21] at $1.2840/1k and 88.40%/97.40%, Hybrid LLM [24] at $1.1920/1k, 1,650.0 ms, and 92.40% Turn-1 / 98.70% Wf, and RouteLLM [22], all calibrated on $N_{\text{cal}} = 250$ to target $Q^* = 0.95$; note that standalone raw-prompt routers operate without Stage 3's unified-diff directive or `schema_audit` trailers [68.4–72.0 out tok], whereas *MINLP Routing Only (Raw Prompts)* [138.2 out tok, $1.3250/1k] includes the multi-tier `schema_audit` trailer on raw full-file outputs and drops to 71.2 out tok, $1.2190/1k without `schema_audit`, or $1.1480/1k when paired with Stage 3's unified-diff directive), we evaluate RouteLLM both as a standard binary tier selector ($T_{\text{think}} = 0$, reaching 91.20% Turn-1 / 98.60% Wf) and augmented with access to the exact same reasoning-enabled Flash-Thinking action space ($\{(\text{Flash}, T_{\text{think}} \in \{0, 4, 8, 16, 128\}), (\text{Argon}, 0)\}$, matching recent joint model-and-test-time-compute routing paradigms [24], [25]): *RouteLLM + Flash-Thinking (Raw)* ($1.1640/1k, 94.40% Turn-1 / 99.10% Wf) and *Compaction + RouteLLM + Flash-Thinking* ($1.0980/1k with `schema_audit` on routed Flash turns [63.0 out tok], and $1.0128/1k at 1,421.5 ms in core-output-only mode; 96.80% Turn-1 / 99.40% Wf) improve over binary routing but still underperform PromptVM's closed-form AST-cyclicity-aware Eq. (26) allocation ($1.0663/1k with trailer and $0.9812/1k at 1,397.9 ms core-only; 99.70% Turn-1 / 100.00% Wf) because text-embedding routers fail to detect structural schema/import cycles ($\delta_{\text{inst}} = 0.030$) on 3.2% of traces (across all 5 seeds, PromptVM achieves $\$1.0665 \pm \$0.0042/1\text{k}$, $1{,}490.1 \pm 18.4$ ms, and $99.68\% \pm 0.08\%$ Turn-1 vs. *Compaction + RouteLLM + Flash-Thinking* at $\$1.0983 \pm \$0.0051/1\text{k}$, $1{,}513.0 \pm 19.2$ ms, and $96.76\% \pm 0.15\%$ Turn-1, yielding paired per-seed improvements of $\Delta\text{Cost} = -\$0.0318 \pm \$0.0031/1\text{k}$, $\Delta\text{Latency} = -22.9 \pm 4.6$ ms, and

$\Delta$Turn-1 $= +2.92\% \pm 0.12\%$, $p < 0.001$).

Finally, to verify external generalization across public benchmarks with 95% bootstrap confidence intervals (and zero benchmark-specific router recalibration, where Eq. 25 operates as an *ordinal difficulty-margin score* over the inter-tier scaling gap $\Delta Q_{\max}(M)$ rather than requiring cardinal Pass@1 calibration—so uniform domain accuracy shifts $-\Delta_{\text{domain}}$ across both tiers on harder benchmarks such as BIRD-SQL or SWE-bench preserve the relative complexity ordering of $\delta_{\text{cohort}} + \delta_{\text{inst}}(x)$—and where the AST classifier maps SQL tasks zero-shot to Cohort A $[\delta_{\text{A}} = 0.000,\ \hat{L}_{\text{out,A}} = 64]$, function-calling tasks zero-shot to Cohort B $[\delta_{\text{B}} = 0.015,\ \hat{L}_{\text{out,B}} = 96]$, Python repository patches in SWE-bench Verified zero-shot to Cohort D $[\delta_{\text{D}} = 0.035]$ via Python `def`/`from ... import` AST rules—partitioning $n = 100$ into 38% single-module acyclic patches routed to Flash at $T^*_{\text{think}} = 128$ and 62% multi-module cyclic patches with $\delta_{\text{inst}} = 0.030$ routed to Argon—and RouterBench tasks by domain across Cohort A/B for MMLU $[\delta \in \{0.000, 0.015\}]$, Cohort C for GSM8K math $[\delta_{\text{C}} = 0.020$, plus $\delta_{\text{inst}} = 0.030$ when $\geq 4$ relational equation steps matching `r"(?:[0-9a-zA-Z\)\]]\s*(?:=|<=|>=)\s*-?[0-9a-zA-Z\(\$]|<<[^>]+=[^>]+>>)"` occur], Cohort D for MBPP code $[\delta_{\text{D}} = 0.035]$, and Cohort E for MT-Bench multi-turn dialogues $[\delta_{\text{E}} = 0.010]$), we evaluated PromptVM under strict single-turn pre-escalation inference ($\bar{R}_k = 1.0, p_{\text{esc}} = 0$) across four public benchmarks: (1) on the **full BIRD-SQL Dev** benchmark [13] ($N = 1{,}534$ queries across all 11 databases, alongside our $n = 200$ stratified subset, at $\bar{R}_k = 1.0, p_{\text{esc}} = 0$), Stages 1–3 on fixed Gemini 4.0 Argon achieved $40.8\% \pm 0.9\%$ input token compaction (614.2 $\rightarrow$ 363.6 tokens; 41.2% on $n = 200$), lifted single-turn execution accuracy from $68.4\% \rightarrow 69.1\% \pm 1.2\%$ (68.5% $\rightarrow$ 69.0% on $n = 200$), and reduced cost by 34.5%, while enabling Stage 5 Pareto routing preserved $69.1\% \pm 1.2\%$ execution accuracy and increased total cost savings to **60.9% ± 1.4%** (61.4% on $n = 200$); (2) on the **full Berkeley Function Calling Leaderboard (BFCL)** AST evaluation suite [2] (https://gorilla.cs.berkeley.edu/leaderboard.html, $N = 2{,}000$ test cases across simple, multiple, parallel, and parallel-multiple AST categories at $\bar{R}_k = 1.0, p_{\text{esc}} = 0$), Stages 1–3 on fixed Gemini 3.8 Flash achieved $38.1\% \pm 0.7\%$ token compaction (509.5 $\rightarrow$ 315.4 tokens; 38.4% on $n = 200$), improved single-turn AST accuracy from $94.6\% \rightarrow 95.1\% \pm 0.6\%$, and reduced cost by 36.4%, while enabling Stage 5 dynamic thinking allocation lifted AST accuracy to **95.5% ± 0.5%** with zero regressions; (3) on a stratified $n = 100$ repository-level code-repair subset of **SWE-bench Verified** (where both baseline Argon and PromptVM emit standard unified-diff patches, so savings reflect pure compaction and Flash/Argon routing), PromptVM's AST cyclicity classifier and routing reduced single-turn inference cost ($\bar{R}_k = 1.0, p_{\text{esc}} = 0$) by **29.4% ± 1.6%** while preserving 100% of baseline single-turn resolved-issue Pass@1 (54.0% $\rightarrow$ 54.0%, lifting post-verification resolution to 58.0% with a single pytest sandbox escalation; ablating $\delta_{\text{inst}}(x) \equiv 0$ misroutes multi-module cyclic patches to Flash, dropping single-turn Pass@1 to 51.0% or reducing quality-matched cost savings to $26.8\% \pm 1.5\%$); and (4) on the public **RouterBench** benchmark [26] ($N = 1{,}000$ stratified prompts across MMLU, GSM8K, MBPP, and MT-Bench at $\bar{R}_k = 1.0, p_{\text{esc}} = 0$ and target quality $Q^* = 0.95$), Stage 5 reduced total inference cost by **34.2% ± 1.1%** relative to FrugalGPT and **14.8% ± 0.8%** relative to RouteLLM while maintaining $95.3\% \pm 0.7\%$ Pass@1 (whereas ablating $\delta_{\text{inst}}(x) \equiv 0$ drops Pass@1 to $94.1\% \pm 0.8\%$ and reduces quality-matched cost savings to $31.8\% \pm 1.2\%$ vs. FrugalGPT and $12.6\% \pm 0.9\%$ vs. RouteLLM).

## VI. Related Work

**Prompt and Tool-Schema Compression:** Prior prompt compression works such as LLMLingua [17], LongLLMLingua [18], and Selective Context [20] filter tokens with low surprisal under small language models, while LLMLingua-2 [19] introduces task-agnostic data distillation. However, token-level statistical filtering frequently prunes negative polarity constraints, quotation delimiters, and numerical bounds, causing functional failures in structured tool and enterprise database tasks (such as BIRD-SQL [13] and BFCL [2]). While deterministic RFC 8259 JSON whitespace minification [8], constrained grammar execution frameworks [11], and unguarded schema-description stripping reduce surface character counts, unguarded stripping discards sibling parameter roles and domain units, whereas embedding-based tool retrieval pipelines [2] risk recall failures on multi-hop compositional tool chains. Going beyond format-level minification or lossy schema retrieval, PromptVM's Stage 1 performs lossless contract-preserving AST canonicalization, 4-part semantic role/unit guarding, and Unigram/BPE subword boundary-guarded `$defs`/`$ref` sub-tree hoisting across all active schemas while Stage 2 guarantees polarity-aware predicate retention.

**Attention Sinks, Modular KV-Cache Reuse, and Test-Time Compute Routing:** StreamingLLM [14] demonstrated that initial sequence tokens act as natural attention sinks in

TABLE III
System Component Ablation Matrix and Baseline Comparison
($N = 1{,}000$, 5 Seeds, Median Seed $s = 42$)

| System Configuration | In Tok | Out / $\bar{R}_k$ | Cost/1k | Latency | Turn-1 / Wf Pass[†] |
|---|---|---|---|---|---|
| Monolithic Baseline (Gemini 4.0 Argon) | 466.7 | 74.3 / 1.55 | $1.7608 | 2,450.0 ms | 85.00% / 98.50% |
| Monolithic Baseline (Diff-Matched Cohort D, Argon) | 466.7 | 51.9 / 1.52 | $1.4568 | 2,198.4 ms | 85.40% / 98.50% |
| Deterministic Schema Minification (RFC 8259 [8])[§] | 416.3 | 69.8 / 1.53 | $1.6980 ($1.6120) | 2,320.0 ms | 85.00% / 98.50% |
| Unguarded Schema Description Stripping (Argon) | 392.1 | 58.6 / 1.54 | $1.5910 | 2,185.0 ms | 79.50% / 91.40% |
| Stage 1a Only (Key Sort + Union Fold + Guard, Argon)[§] | 411.2 | 66.2 / 1.51 | $1.6915 ($1.5780) | 2,255.0 ms | 85.00% / 98.50% |
| Stage 1 Complete (1a + `$defs`/`$ref` Hoisting, Argon)[§] | 408.5 | 65.4 / 1.51 | $1.6881 ($1.5642) | 2,240.0 ms | 85.00% / 98.50% |
| Stage 2 Only (Span-Strip + Invariant Pruning, Argon)[§] | 440.6 | 62.1 / 1.50 | $1.7282 ($1.5180) | 2,190.0 ms | 82.10% / 95.80% |
| LLMLingua Compression (Full Prompt, Argon) [17] | 350.0 | 54.2 / 1.62 | $1.4700 | 2,120.0 ms | 64.50% / 74.20% |
| LLMLingua-2 Distillation (Full Prompt, Argon) [19] | 342.0 | 53.0 / 1.58 | $1.4480 | 2,080.0 ms | 71.80% / 82.40% |
| LLMLingua-2 (Matched Budget = 384.9 tok, Argon) [19] | 384.9 | 56.4 / 1.56 | $1.5420 | 2,140.0 ms | 75.40% / 86.10% |
| Selective Context (Full Prompt, Argon) [20] | 373.3 | 55.8 / 1.60 | $1.5280 | 2,190.0 ms | 68.20% / 78.50% |
| Schema-Safe LLMLingua ($I$ Only) + Stage 1 AST | 398.2 | 58.1 / 1.48 | $1.5840 | 2,040.0 ms | 81.20% / 94.60% |
| FrugalGPT Cascade [21] | 466.7 | 72.0 / 1.49 | $1.2840 | 2,150.0 ms | 88.40% / 97.40% |
| Hybrid LLM Quality-Aware Router (Raw) [24] | 466.7 | 70.8 / 1.44 | $1.1920 | 1,650.0 ms | 92.40% / 98.70% |
| RouteLLM Preference Router (Raw, $T_{\text{think}} = 0$) [22] | 466.7 | 71.5 / 1.45 | $1.2180 | 1,680.0 ms | 91.20% / 98.60% |
| RouteLLM + Flash-Thinking (Raw) [22] | 466.7 | 68.4 / 1.42 | $1.1640 | 1,615.0 ms | 94.40% / 99.10% |
| Compaction + RouteLLM ($T_{\text{think}} = 0$) [22] | 384.9 | 64.2 / 1.43 | $1.1120 | 1,540.0 ms | 94.80% / 98.80% |
| Compaction + RouteLLM + Flash-Thinking [22] | 384.9 | 63.0 / 1.41 | $1.0980 ($1.0128[‡]) | 1,512.0 ms | 96.80% / 99.40% |
| Compaction Only (on Gemini 4.0 Argon)[§] | 384.9 | 42.7 / 1.40 | $1.6586 ($1.2946) | 1,980.0 ms | 100.00% / 100.00% |
| Compaction w/o Landmark Anchors (on Argon)[§] | 382.4 | 42.1 / 1.48 | $1.6554 ($1.3520) | 1,970.0 ms | 82.10% / 95.80% |
| MINLP Routing Only (Raw Prompts) | 466.7 | 138.2 / 1.43 | $1.3250 ($1.2190[‡]) | 1,720.0 ms | 94.20% / 98.00% |
| **PromptVM Full Pipeline (w/ Audit Trailer)** | **384.9** | **134.7 wf / 1.40** | **$1.0663** | **1,489.4 ms** | **99.70% / 100.00%** |
| **PromptVM Full Pipeline (Core Output Only)** | **384.9** | **93.2 wf / 1.40** | **$0.9812** | **1,397.9 ms** | **99.70% / 100.00%** |

[†] Turn-1 Single-Call Pass@1 (Pre-Retry) / Post-Verification Workflow Completion Rate (Wf) under identical sandbox oracles.
[§] Output-normalized cost at fixed Baseline workflow output (74.3 tok wf, $\bar{R}_k = 1.55$), followed by realized cost; [‡] core-only cost with `schema_audit` disabled.

autoregressive models. PromptVM establishes the complementary compile-time principle: synthetically injecting uppercase structural landmark anchors bounds background attention dispersion while co-locating invariant clauses to amplify retrieval-head attention. In runtime memory management, vLLM [10] and SGLang [11] manage PagedAttention block tables and RadixAttention prefix reuse, while modular KV-cache reuse frameworks such as Prompt Cache [23] precompute reusable attention blocks with decoupled positional encodings across document chunks; PromptVM provides the hardware-aligned complement that pads static prefixes to exact HBM2e page boundaries ($B \in \{16, 32\}$) with zero-initialized VMEM register masking and decoupled logical RoPE/sliding-window counters to eliminate copy-on-write boundary-block duplication across concurrent subagents. Finally, in model routing and test-time compute scaling, systems such as FrugalGPT [21], Hybrid LLM [24], RouteLLM [22], and unified routing evaluation suites such as RouterBench [26] explore reactive post-generation cascading and preference-based tier selection, while difficulty-aware test-time compute scaling studies [25] show that allocating variable reasoning compute often outperforms scaling model parameters. Rather than incurring two sequential model invocations via reactive POMDP cascades or sampling multiple parallel completions, PromptVM unifies these paradigms via a single-pass pre-dispatch closed-form discrete MINLP that jointly selects model serving tiers and integer test-time reasoning budgets ($T^*_{\text{think}}$) co-designed with upstream prompt compaction.

## VII. Conclusion

We presented the *Tokenomics Compiler* (`PromptVM`), a hardware-aware compilation and discrete-choice economic routing framework for enterprise LLM serving. By uniting tokenizer-aware AST transpilation, invariant density compaction, attention-sink landmark anchoring, server-side Pallas paged-cache boundary sealing, and discrete Pareto routing, our system slashes per-cohort enterprise inference costs by 26.26% to 88.79% (39.44% pooled invoice savings, 40.22% unweighted mean cohort savings, 44.28% savings without audit trailers, and 26.81% against a diff-matched baseline), eliminates multi-branch copy-on-write boundary page duplication on Cloud TPU v6e, and achieves 99.70% strict Turn-1 Single-Call Pass@1 and 100.00% workflow completion while provably bounding attention dispersion.

*Limitations and Trade-offs:* First, compaction efficiency is bounded by the ratio of structured schemas and boilerplate to dense payload content; on raw source code refactoring (Cohort D), AST schema compression yielded a 1.11% token reduction, with cost savings driven primarily by Stage 5 dynamic reasoning allocation and unified-diff output structuring (validated externally on SWE-bench Verified). Second, achieving $100.00\%$ post-verification workflow completion (lifting $99.70\%$ Turn-1 Pass@1 across the $0.30\%$ residual Cohort D failures) relies on deterministic execution-verification sandboxes (`go build`, Cloud Spanner AST parsers, JSON Schema validators) that add $482$–$714$ ms of verification latency; in open-ended natural language generation tasks lacking an executable oracle, workflow accuracy is governed by the pre-escalation Turn-1 rate ($99.70\%$). Third, while direct white-box attention entropy verification ($\mathcal{H}_{\text{empirical}} \leq \mathcal{H}_{\max}$) is confirmed across open-weights SentencePiece Unigram (Gemma-2-9B) and BPE (Llama-3.1-8B-Instruct) models, proprietary frontier endpoints (Regime A) do not expose internal attention weights, so landmark sink efficacy on closed APIs is validated indirectly via downstream Turn-1 Pass@1 gains ($+17.90\%$ over unanchored compaction in Table III). Fourth, the Stage 5 quality surface is calibrated on $N_{\text{cal}} = 250$ held-out traces using a shared binary cyclicity step penalty $\delta_{\text{inst}} \in \{0, 0.030\}$; while effective across Cohorts B–E, continuous AST complexity features (such as SQL join/recursive-CTE depth in Cohort A write-path DDL/DML or continuous cyclicity depth $C_{\text{task}}(P)$ implemented in our router API) provide finer-grained intermediate thinking budgets, under distribution shift to unseen task domains the router relies on its conservative fallback parameters ($\delta_{\text{default}} = 0.035$, $\hat{L}_{\text{out,default}} = 256$, $\hat{L}_{\text{sandbox,default}} = 600.0$ ms, $L_{\max,\text{default}} = 2{,}000$ ms), and Stage 2's static 64-token lexicon requires domain-specific extension via its configurable `custom_lexicon` interface when new domain-specific invariant markers or unmodalized prose action verbs arise. Fifth, Stages 4 and 5 exhibit an inverse infrastructure trade-off between managed cloud APIs and self-hosted accelerators: while Stages 1–3 and Stage 5 deploy transparently as a client-side SDK middleware on managed elastic cloud APIs (Vertex AI, where multi-tier routing across Flash and Argon incurs zero client memory overhead), Stage 4's interior page padding requires our custom server-side JAX/Pallas kernel on self-hosted Cloud TPU v6e clusters (where our $14.23\%$ 8-way TTFT reduction was benchmarked on a single TPU v6e chip hosting Gemma-2-9B at $B = 16$, whereas frontier multi-chip tensor- and sequence-parallel pod deployments shard KV-cache pages across multiple HBM spaces and incur additional Inter-Chip Interconnect [ICI] collective latencies during prefill); conversely, operating Stage 5's multi-tier routing on a fixed-capacity self-hosted TPU pod requires partitioning HBM across multiple model weights ($\{\text{Flash}, \text{Argon}\}$), trading per-tier KV-cache capacity against routing elasticity. Future work will extend AST transpilation to programming language syntax trees via Tree-sitter parsers and integrate multi-tier host DRAM-to-HBM speculative page prefetching.

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